Signals & Systems

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Course Information

Text Book:

Signals & Systems, Second Edition by Oppenheim

Contents: Chapter 1 — Chapter 10

Reference: "Signals and Systems", Junli Zheng

TA: Tengyu Zhang (zhang-ty22@mails.tsinghua.edu.cn)

Homework DDL: Due every Tuesday after a Chapter is completed

Q&A: Every Fri. 17:05-18:05 PM, venue: Main Building 627D

Trick: Reading the text book is very important

Grade Components

- Class participation (10%)
  - Attend and be active for each lecture (5%)
  - Feedback on tricky problems and how to improve the lecture will be appreciated (5%)
- Homework (20%)
  - 10 assignments
  - No collaboration
- 3 Quiz (15%)
- 1 Final (55%)

What is the information (message)?

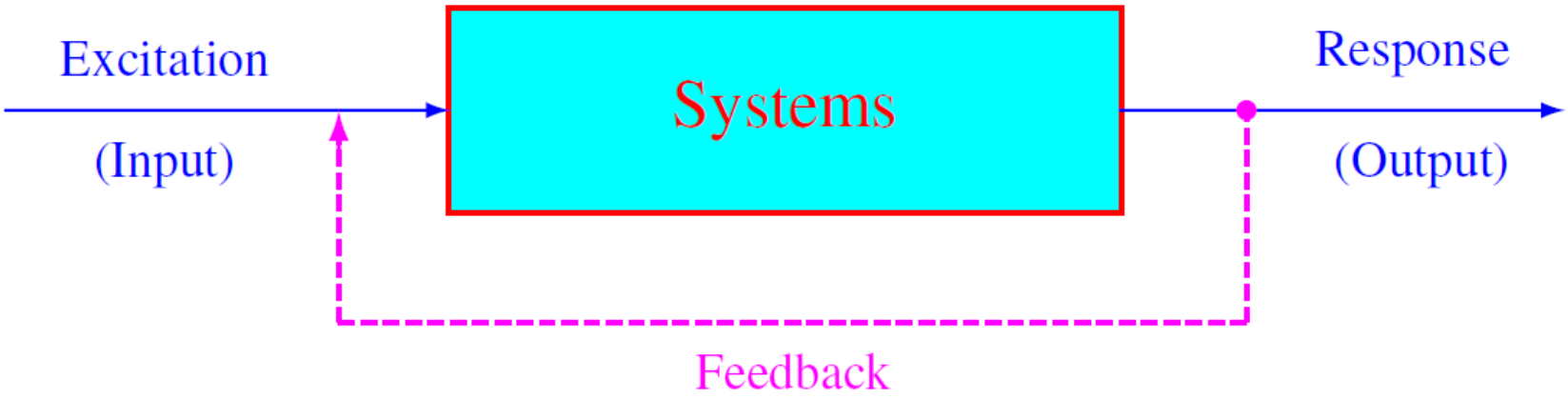
- Information is the context in which data is taken. —— Wikipedia
- (Something which gives) knowledge in the form of facts, news, etc. ——Longman Dictionary of Contemporary English
- 对于消息接受者来说预先不知道的报道——《辞海》
- Information is negative entropy ——C. E. Shannon

Relationship between signal and information:
Signal——The carrier of information
Information——The content of signal

What Are Signals and Systems?

- The concepts of signals and systems arise in a wide variety of fields
- A signal is any physical phenomenon which conveys **information (message)**, it can be represented by a function of one or more independent variables.
- Systems respond to signals and produce new signals
  - Circuit
  - Car
  - Yourself
- Excitation signals are applied at system inputs and response signals are produced at system outputs

- Excitation signals are applied at system inputs and response signals are produced at system outputs



- Example:
  - Current and Voltage are functions of time: signal
  - Circuit: system

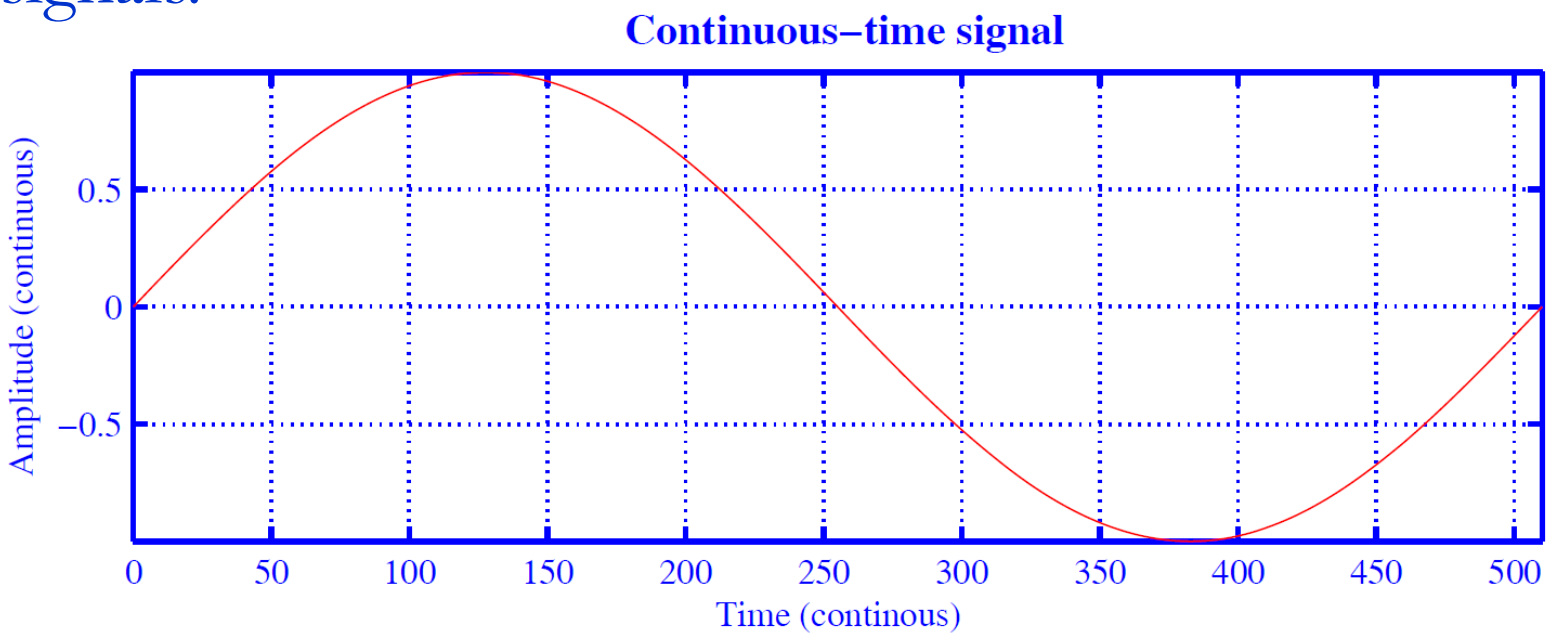
Technical Terms

- Signal Processing: the analysis, interpretation and manipulation of acquired signals so to achieve certain purpose.
  - Signal analysis: analysis of the characteristics in signal, e.g., the frequency and the periodicity of the signal.
  - Signal design: design signals with certain feature. e.g., design the signal shape to reduce the inter-symbol-interference(ISI)
  - System analysis: how the system responds to various input signals, e.g., circuit analysis to different voltage
  - System design: design systems such that the output would meet certain requirement, e.g., high signal-to-noise ratio (SNR)

Design after analysis

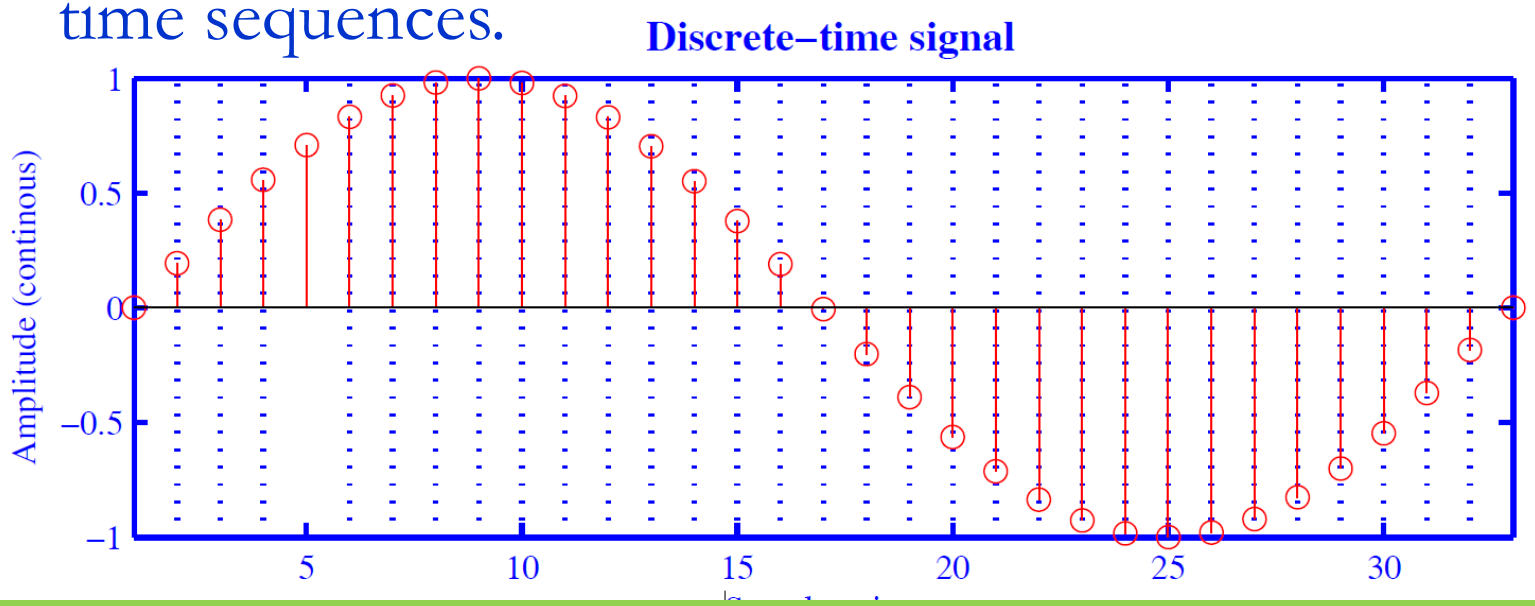
Continuous-Time Signals

- A continuous-time signal is defined along a continuum of times, and thus is represented by a continuous independent variable, $x(t)$
- Continuous-time signals are often referred to as analog signals.



Discrete-Time Signals

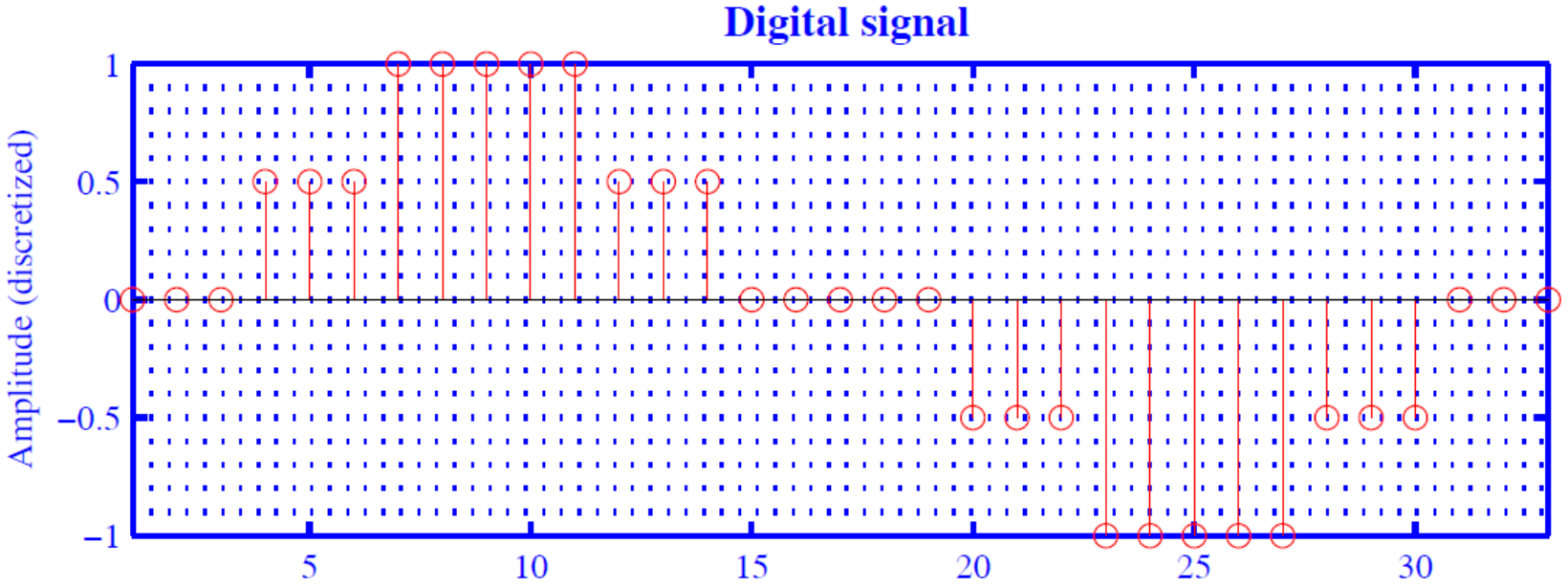
- A discrete-time signal is defined at discrete times, and is represented as sequences of numbers, $x[n]$, whose domain is a set of integers
- Discrete-time signals are often referred to as discrete-time sequences.



Usually, discrete-time signals are obtained by sampling the continuous-time signals

Digital Signals

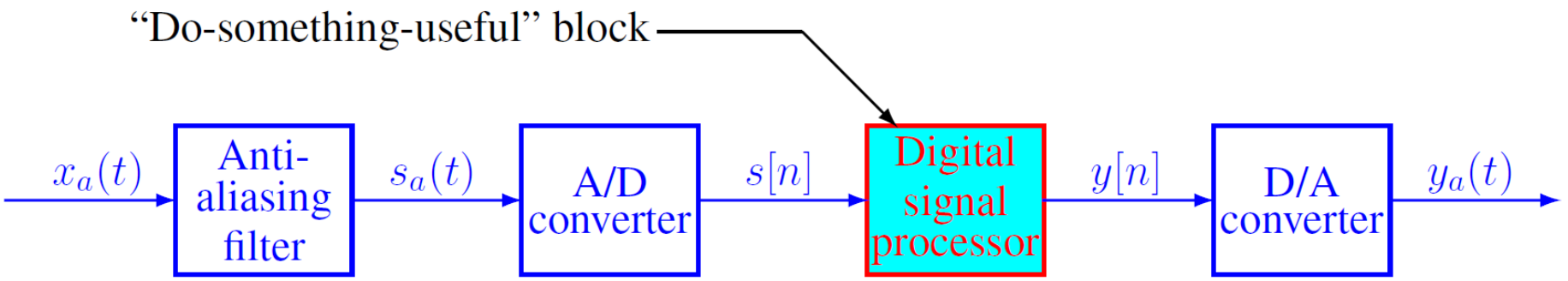
- Digital signals are those for which both time and amplitude are discrete.
- Usually, the digital signal is obtained by an analog-to-digital converter (ADC): Quantization.



How about continuous signal with discrete amplitude?

Related Signal Processing

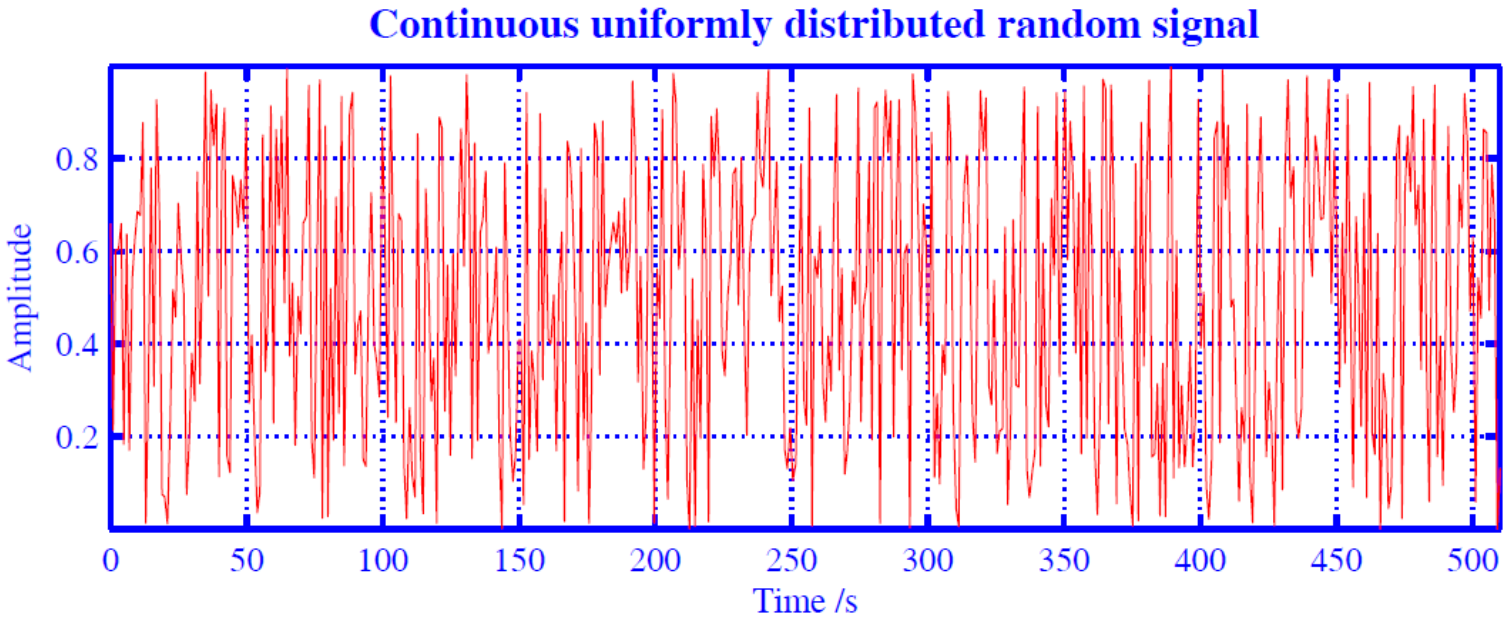
- Fundamental, concept and method are essential for the further study in the areas of communications & control. (with explicit signal modeling)



- Also very useful for image processing, computer vision, and machine learning, etc. (without explicit signal modeling)

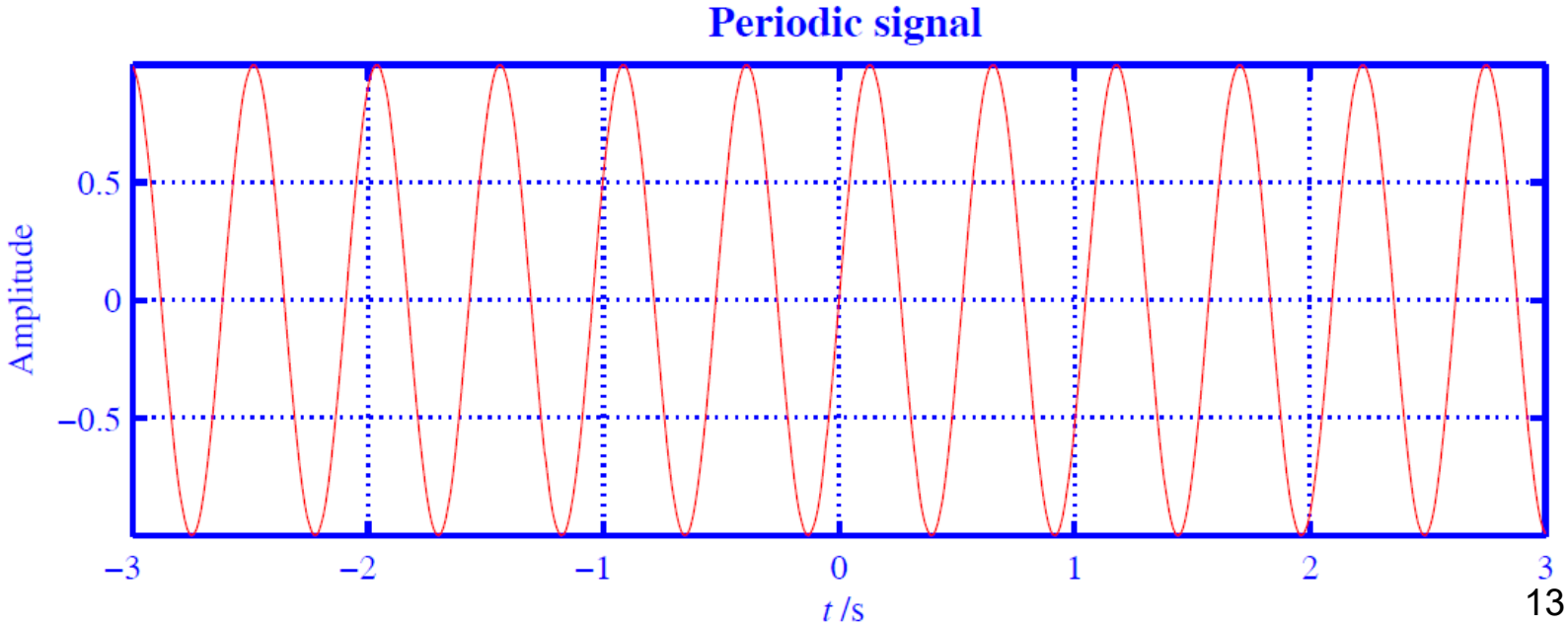
Deterministic and Random Signals

- Deterministic signals: functions have deterministic values at every time and can be determined by a mathematical expression, rule, or table, $e^{j2\pi f_0 t}, \sin 2\pi f_0 n$
- Random signals: A signal has a lot of uncertainty about its behavior at every time slot, $\sin(2\pi f_0 t + \boldsymbol{\theta})$



Periodic and Aperiodic Signals

- Periodic signals repeat with some period T (for continuous-time case) or N (for discrete-time case), while aperiodic (or nonperiodic) signals do not
  - $x(t) = x(t+T), \quad T > 0 \;, \quad x[n] = x[n+N], \quad N > 0$
  - The smallest period is defined as the fundamental period.



What happens when $T \to +\infty$

How about constant signals?

Signal Energy and Power

- Signals directly related to energy and power, (e.g. circuit)
  - Energy:

$$E = \int_{t_1}^{t_2} |x(t)|^2 dt \qquad E = \sum_{n=n_1}^{n_2} |x[n]|^2$$

Complex number x in this course.
How?

  - (average) Power:

$$P = \frac{1}{t_2 - t_1} \int_{t_1}^{t_2} |x(t)|^2 dt \qquad P = \frac{1}{n_2 - n_1 + 1} \sum_{n_1}^{n_2} |x[n]|^2$$

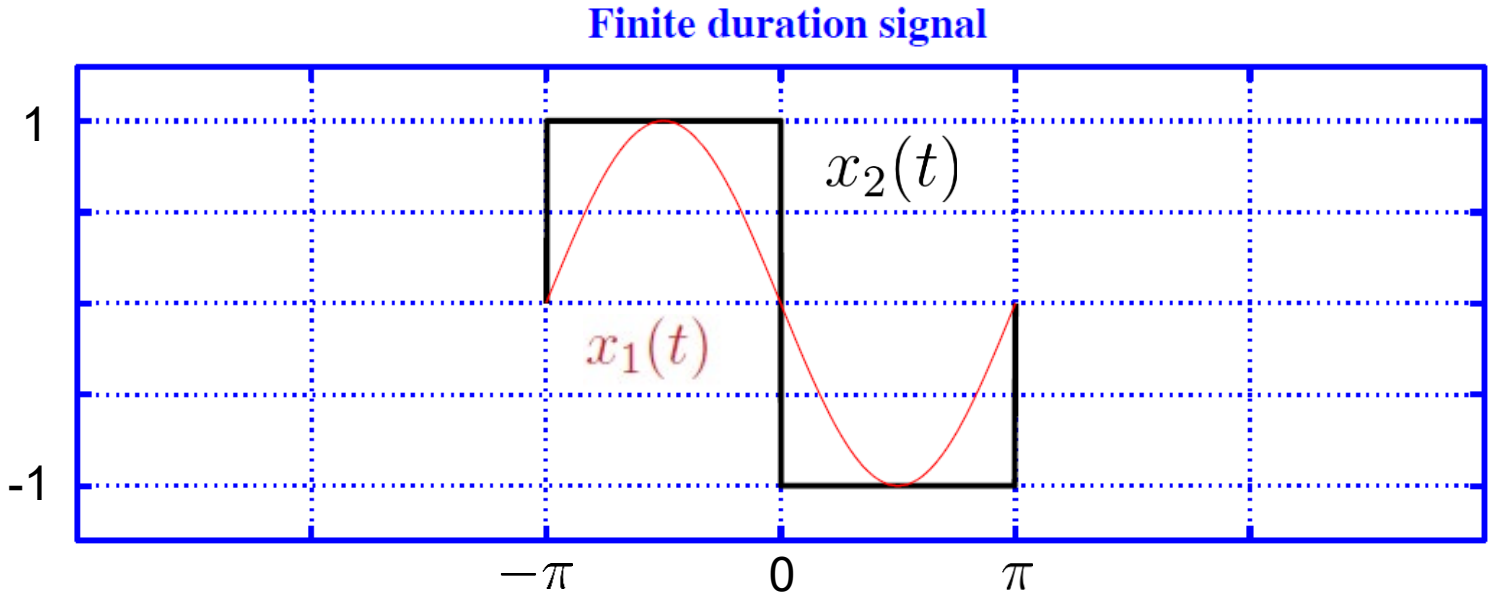
  - For infinite interval

$$E_\infty = \int_{-\infty}^{+\infty} |x(t)|^2 dt \qquad E_\infty = \sum_{-\infty}^{+\infty} |x[n]|^2$$

$$P_\infty = \lim_{T\to\infty} \frac{1}{2T} \int_{-T}^{T} |x(t)|^2 dt \qquad P_\infty = \lim_{N\to\infty} \frac{1}{2N+1} \sum_{-N}^{N} |x[n]|^2$$

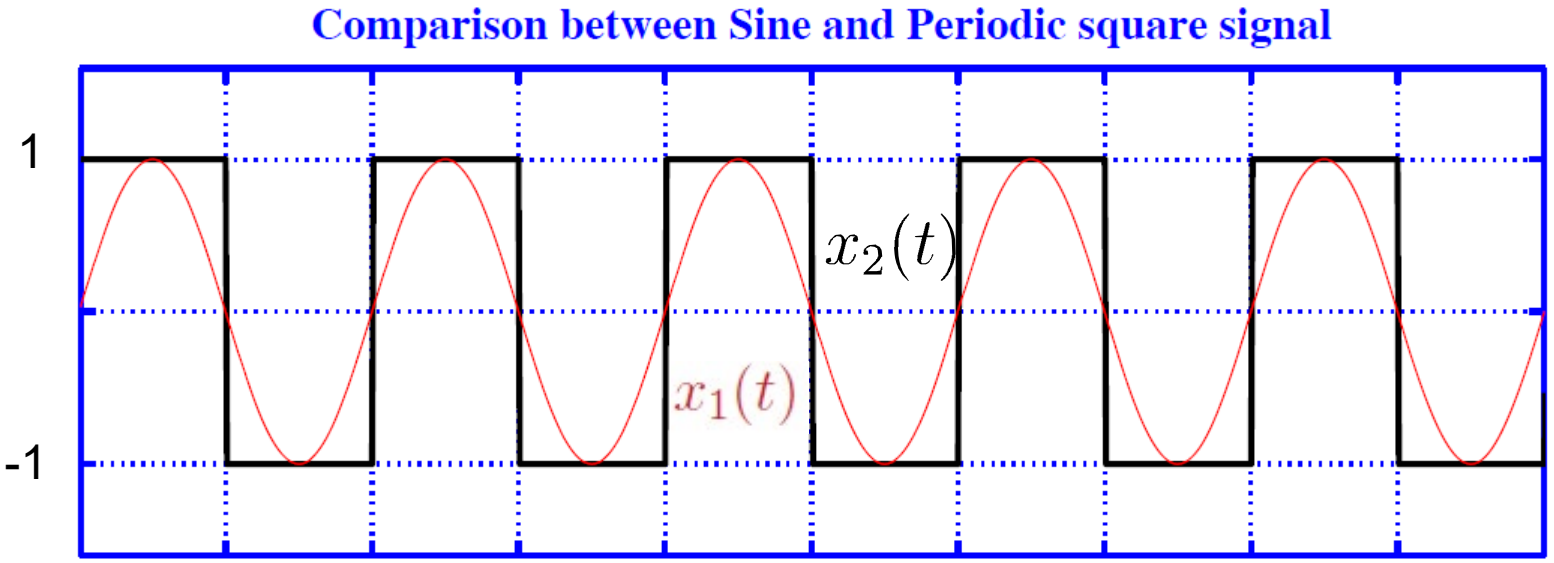
- With above definitions:
  - Energy signals——have finite energy ($E_\infty < \infty$)
  - Power signals——have finite power ($P_\infty < \infty$)
  - Neither finite energy nor finite power signals
- Questions?
  - Are all energy signals also power signals?
  - Are all power signals also energy signals?
  - Are all signals either energy or power signals?

- Example: Signals with finite energy



  - $E_1 = \int_0^{2\pi} |\sin(t)|^2 dt = \pi, \quad E_2 = \int_0^{\pi} |1|^2 dt + \int_{\pi}^{2\pi} |-1|^2 dt = 2\pi$
  - $P_1 = P_2 = 0$, power is not suitable to compare the two signals

- Example: Signals with finite power

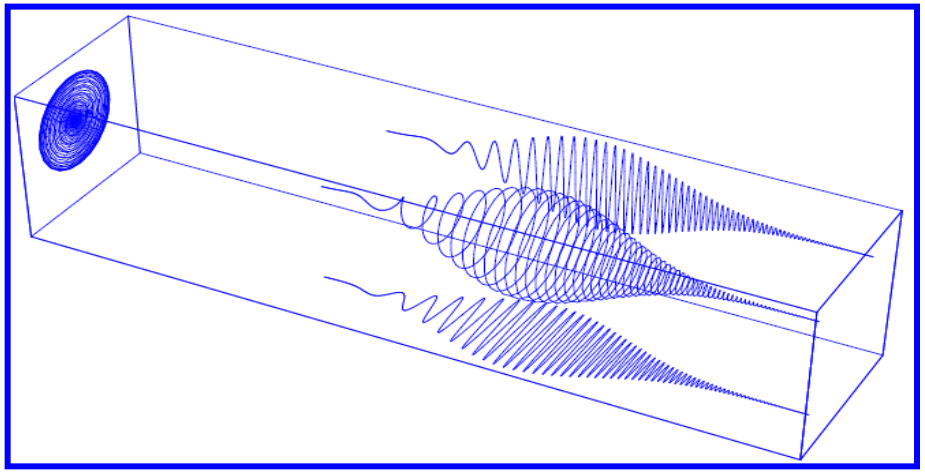


  - $P_1 = \frac{1}{2\pi}\int_0^{2\pi} |\sin(t)|^2 dt = \frac{1}{2}, \quad P_2 = 1$

  - $E_1 = E_2 = \infty$, energy is not suitable to compare the two signals

One- and Multi-Dimensional Signals

- This classification of signals is based on the number of independent variables (i.e., dimensionality).
  - One-dimensional signal (e.g., audio)
  - Multi-dimensional signal (e.g., image)



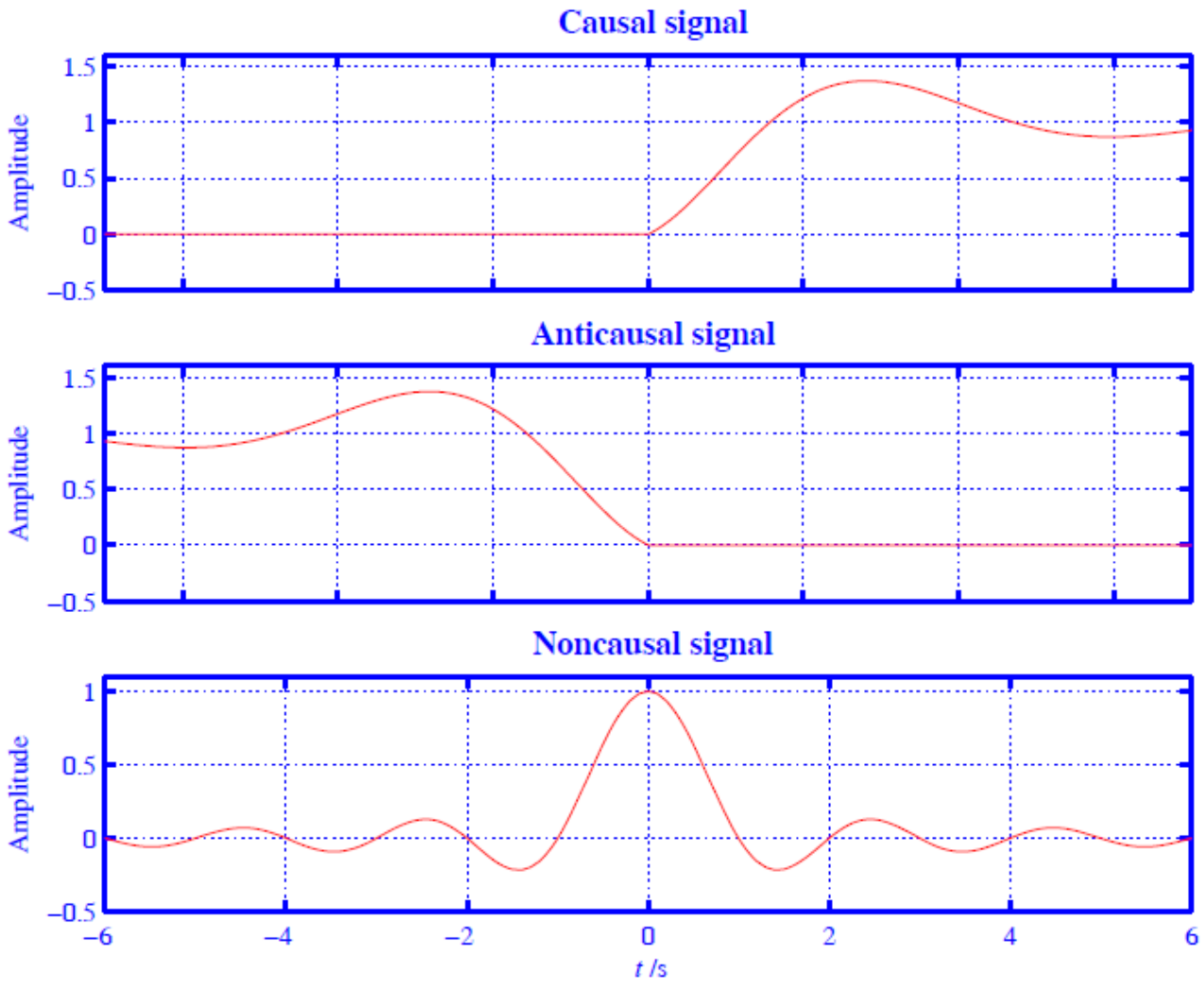


Causal, Anticausal, and Noncausal Signals

$x(t) = 0, \quad t < 0$

$x(t) = 0, \quad t > 0$

Otherwise



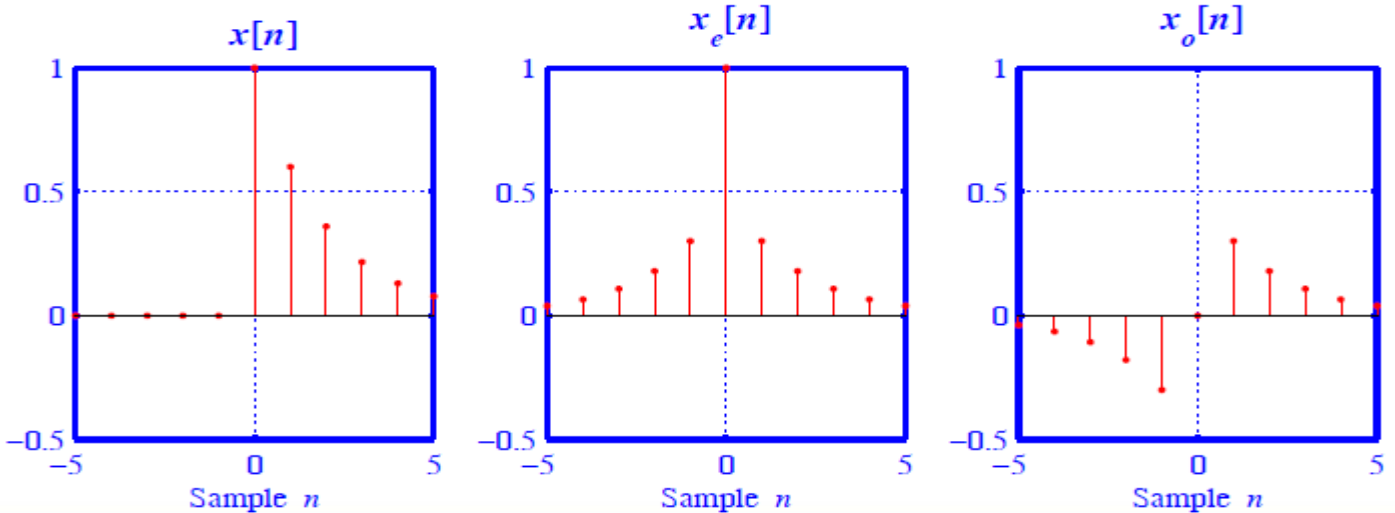
Even and Odd Signals

- Even and odd signals:
  - Even: $x(t) = x(-t) \qquad x[n] = x[-n]$
  - Odd: $x(t) = -x(-t) \qquad x[n] = -x[-n]$

What is $x(0)$?

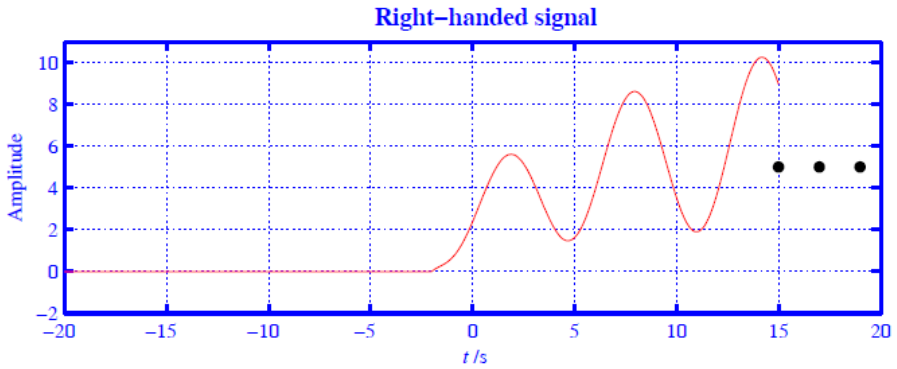
  - Any signal can be represented by a sum of even and odd signals: $x(t) = x_e(t) + x_o(t)$

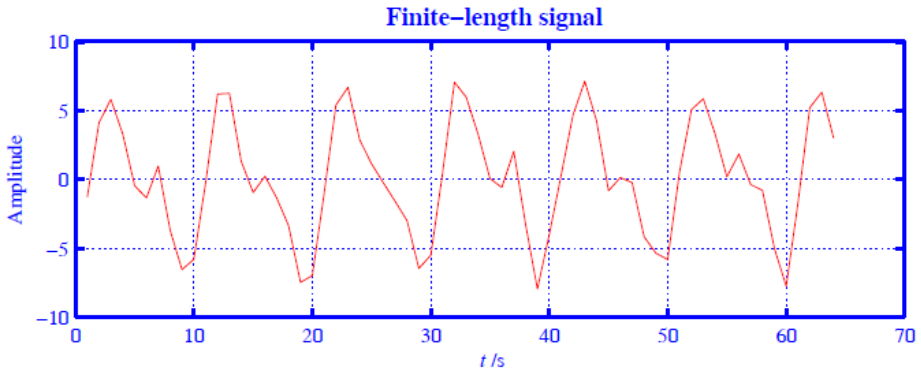
$$x_e(t) = \frac{1}{2}[x(t) + x(-t)] \qquad x_o(t) = \frac{1}{2}[x(t) - x(-t)]$$

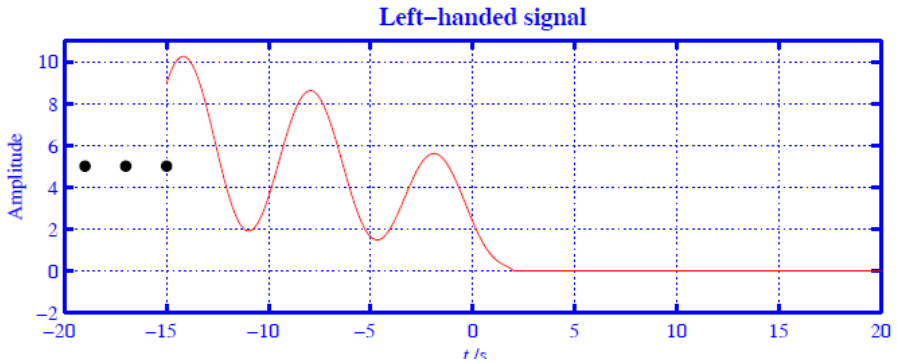


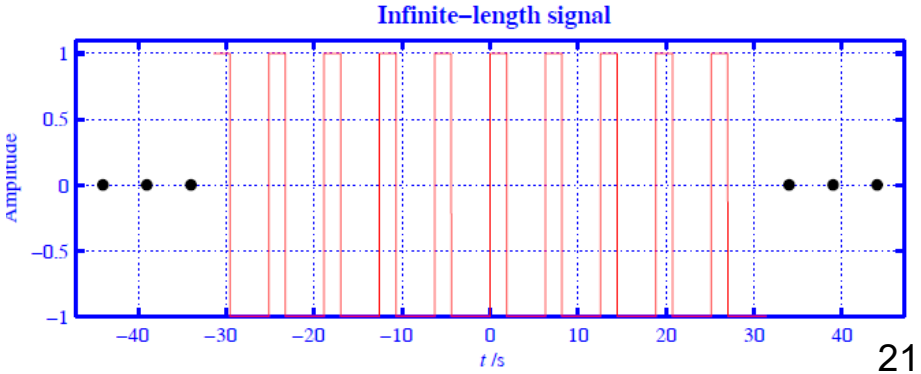
Other Classifications

- Right-Handed & Left-Handed Signals
- Finite & Infinite Length Signals
- ……









Fundamental Operations on Signals

- Addition

$$y(t) = x(t) + w(t)$$

  - Example: wireline communications
- Multiplication

$$y(t) = w(t)x(t)$$

  - Example: Code division multiple access (CDMA)
- Scalar Multiplication

$$y(t) = \alpha x(t)$$

  - Example: wireless communications

Fundamental Operations on Signals

- Delay/Time Shift

  $x(t - t_0),\ x[n - n_0]$

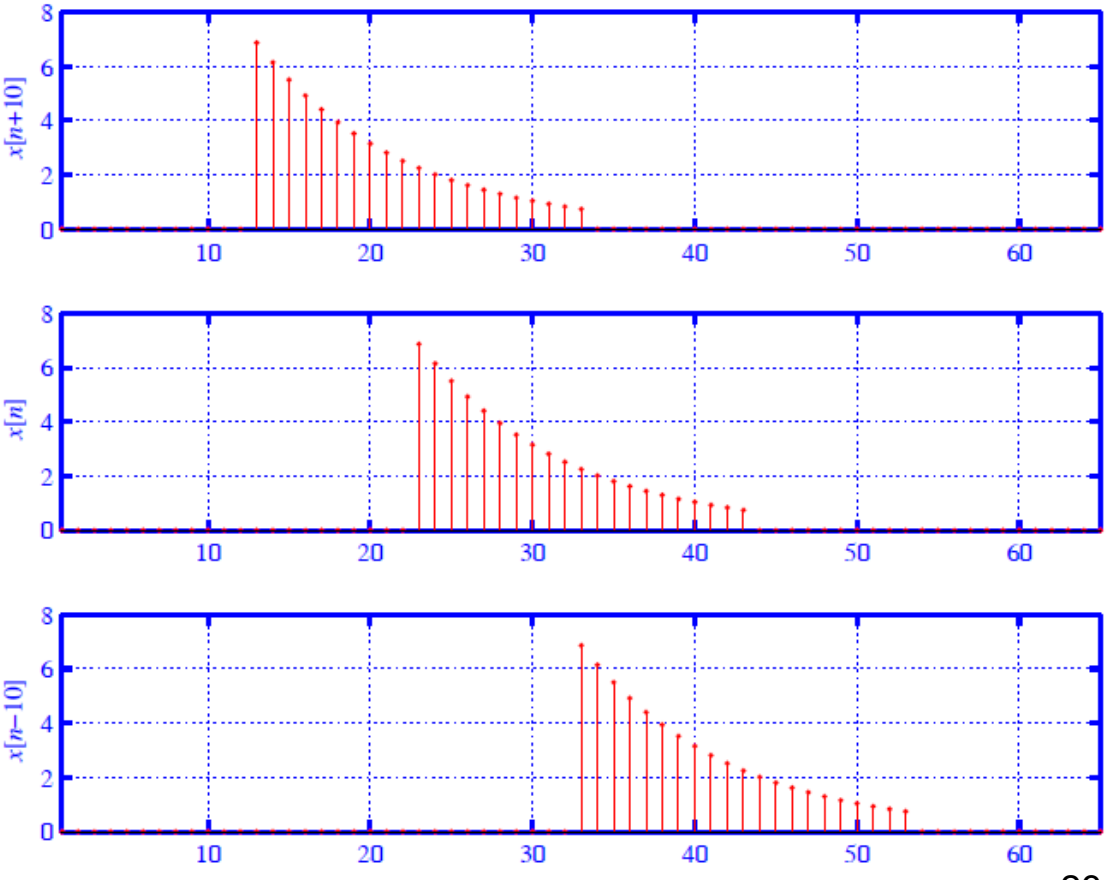
  – To the right (delay)

  $t_0 > 0,\ n_0 > 0$

  – To the left (advance)

  $t_0 < 0,\ n_0 < 0$

  – Delay inherently present in many systems, but may artificially introduced to aid processing



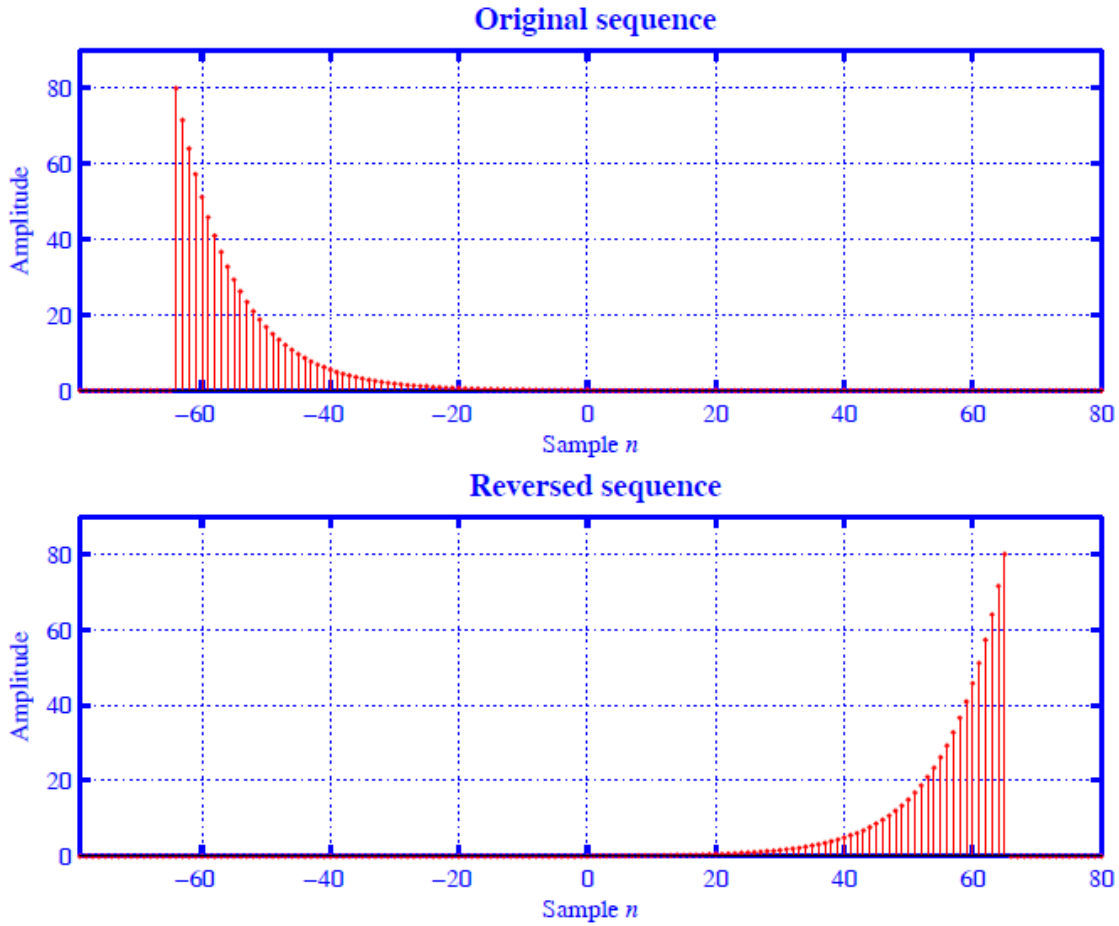
OTA transmission, reflection, echo

Fundamental Operations on Signals

- Time Reversal $y(t) = x(-t),\ y[n] = x[-n]$

反褶

Reverse a tape



In some situations, time reversal operation is combined with the time shifting operation. For example, consider

$$y(t) = x(-(t - t_0)) = x(t_0 - t)$$

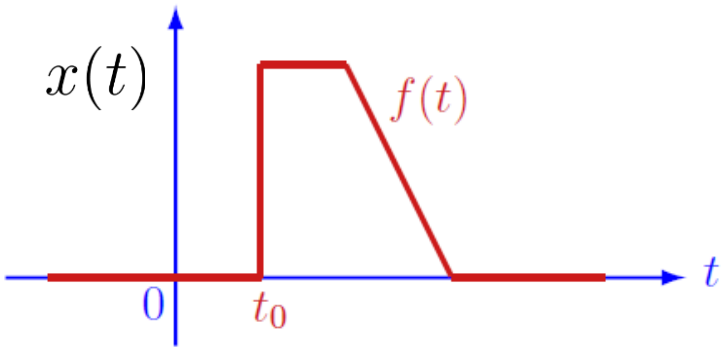
Example

$y(t) = x(2 - t)$

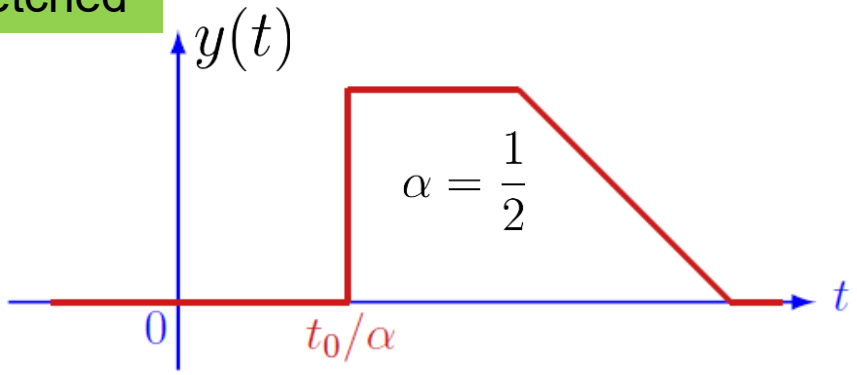
Fundamental Operations on Signals

- Time scaling $y(t) = x(\alpha t),\ \alpha \in \mathbb{R}^+$

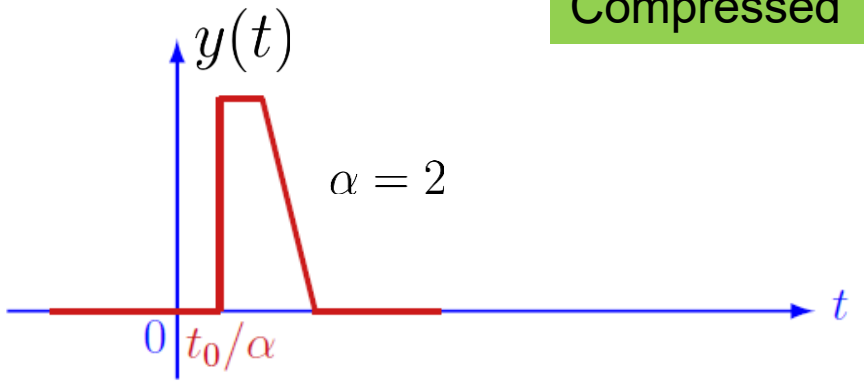
Speed up or slow down the tape



Stretched



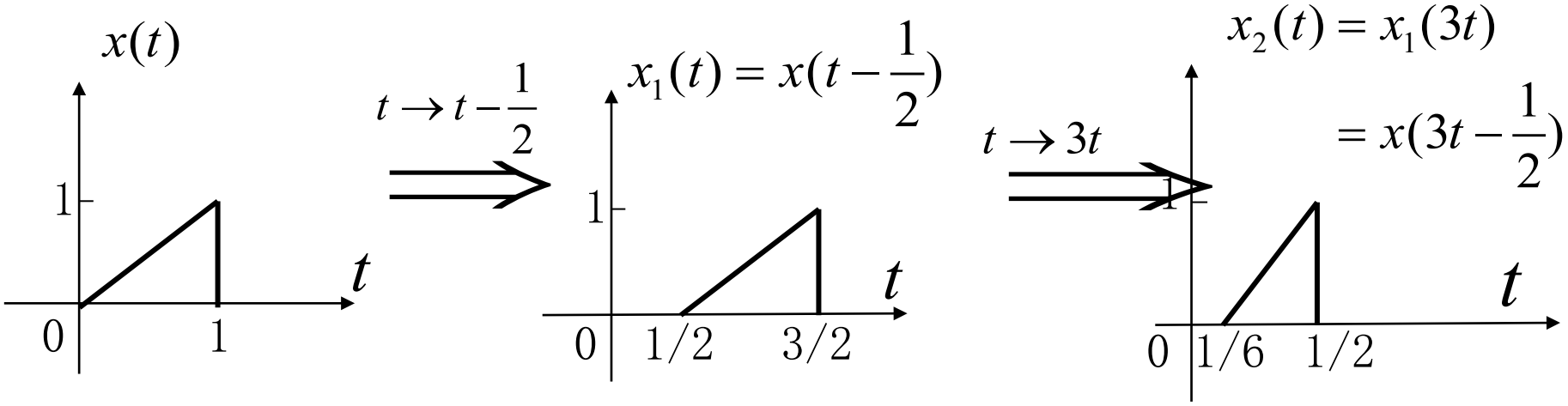
Compressed



- If $\alpha < 0$, there is a time reversal and scaling.

How about discrete signal?

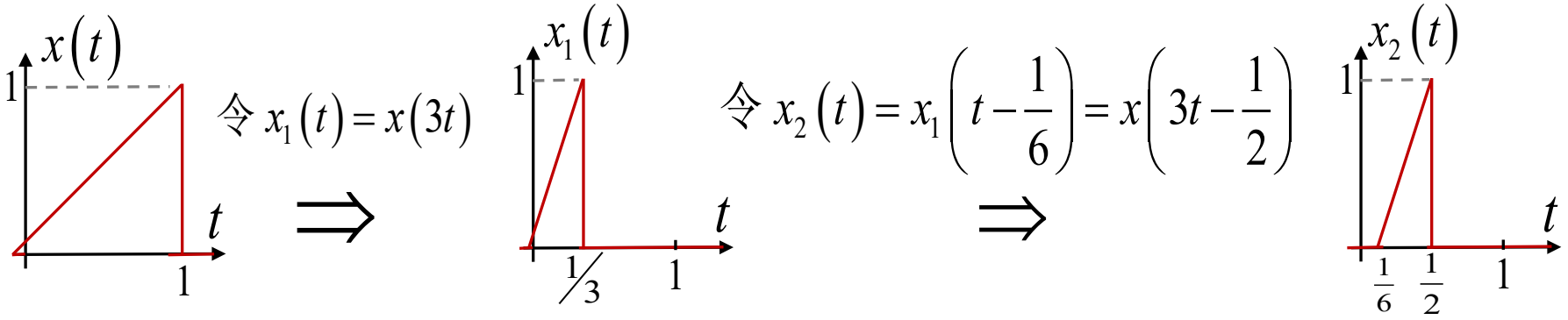
- Example $y(t) = x\left(3t - \frac{1}{2}\right)$



- For $y(t) = x(at + b)$
  - Delay or advance $x(t)$ according to b
  - Perform time scaling/ time reversal according to a
  - Check the zero node or the end node: $x(0), x(1)$

other way?

- The other way:

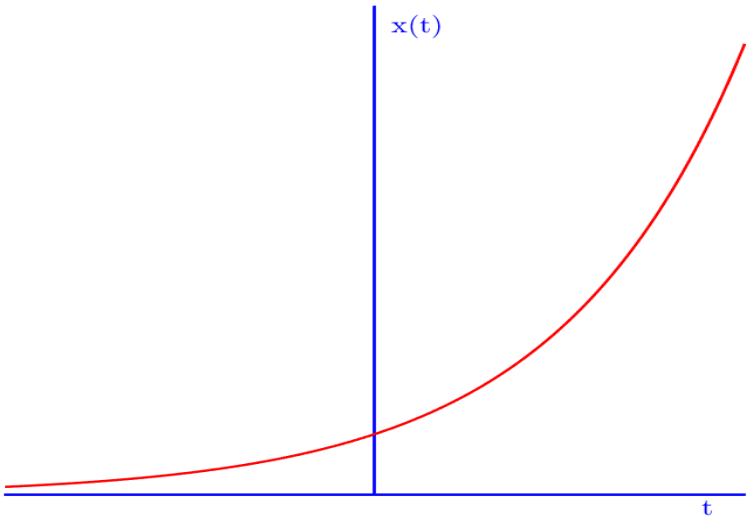


- For
  - Perform time scaling/ time reversal according to a
  - Delay or advance according to b/a
  - Check the zero node or the end node $x(0), x(1)$

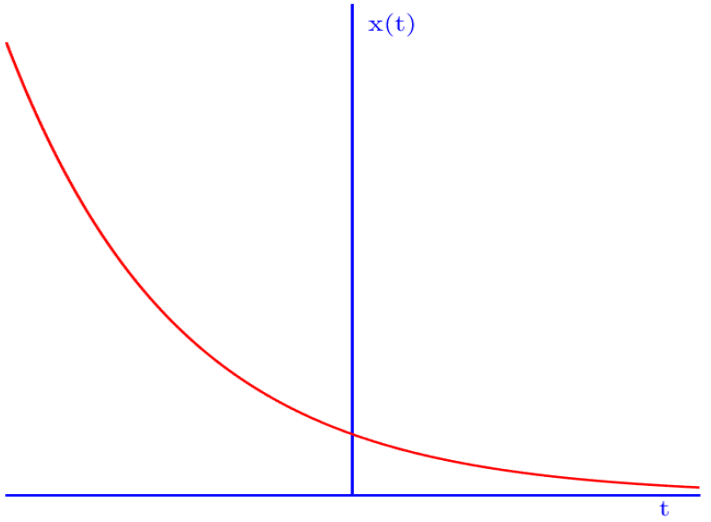
Continuous-Time Exponential and Sinusoidal Signals

- General form: $x(t) = Ce^{at}$
- Real exponential signals: When C and a are real

$a > 0$



$a < 0$



  - How about $a = 0$?

Most physical world signals

Continuous-Time Exponential and Sinusoidal Signals

- Typical complex exponential: $x(t) = e^{j\omega_0 t}$
  - Periodic in T as long as $e^{j\omega_0 T} = 1$
  - The fundamental period is $T_0 = 2\pi/|\omega_0|$
  - The fundamental frequency is $|\omega_0|$
  - Euler's relation: $e^{j\omega_0 t} = \cos\omega_0 t + j\sin\omega_0 t$

Sinusoidal signal with the same fundamental period

$$\cos\omega_0 t = \frac{e^{j\omega_0 t} + e^{-j\omega_0 t}}{2} = \Re\{e^{j\omega_0 t}\}$$

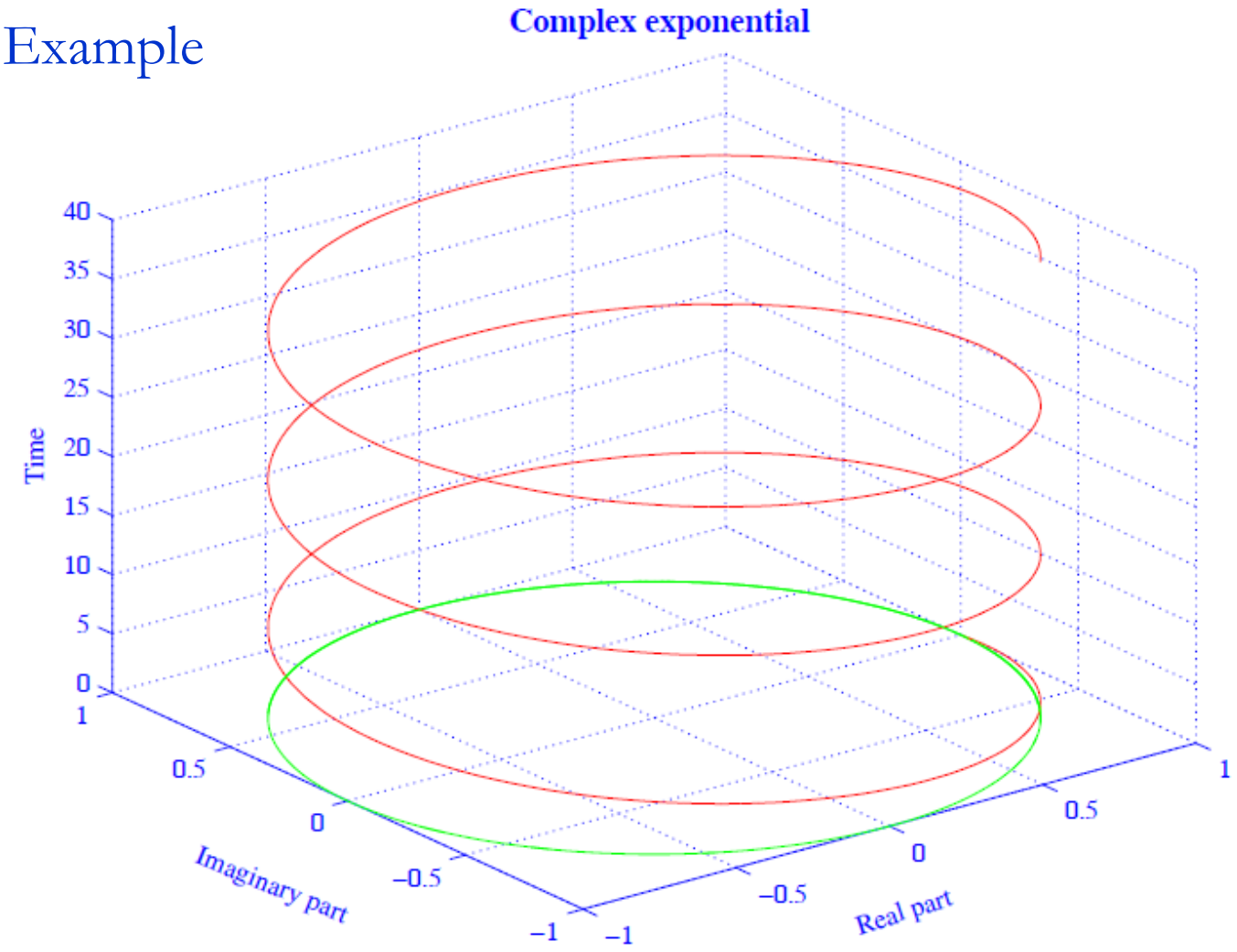
$$\sin\omega_0 t = \frac{e^{j\omega_0 t} - e^{-j\omega_0 t}}{2j} = \Im\{e^{j\omega_0 t}\}$$

$\omega_0 = 2\pi f_0$

  - Sinusoidal signal $x(t) = A\cos(\omega_0 t + \phi)$

$$= \frac{A}{2}e^{j\phi}e^{j\omega_0 t} + \frac{A}{2}e^{-j\phi}e^{-j\omega_0 t} = A\Re\{e^{j(\omega_0 t+\phi)}\}$$

– Example


Complex exponential
Time
40
35
30
25
20
15
10
5
0
1
0.5
0
−0.5
−1
Imaginary part
−1
−0.5
0
0.5
1
Real part

- Energy over one period: $E = \int_0^{T_0} |e^{j\omega_0 t}|^2 dt = \int_0^{T_0} 1 dt = T_0$
- Power over one period is $P = 1$

- Harmonically related complex exponentials---sets of period exponentials that has a common period T_0

$$e^{j\omega T_0} = 1 \quad \Rightarrow \quad \omega = 2k\pi/T_0 = k\omega_0$$

Specifically:

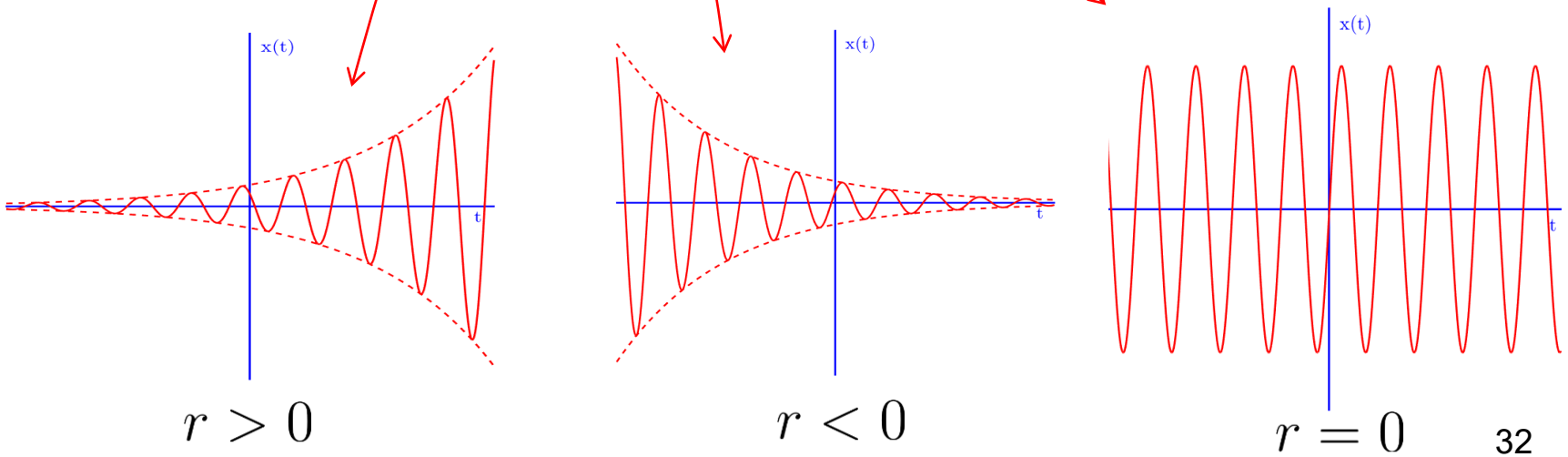
$$\phi_k(t) = e^{jk\omega_0 t}, \quad k = 0, \pm 1, \pm 2, \ldots$$

The k-th harmonic $\phi_k(t)$ is still periodic but with the fundamental frequency $|k|\omega_0$ and fundamental period $T_0/|k|$

- General Complex Exponential signals
  - Let $C = |C|e^{j\theta}, a = r + j\omega_0$, then

$$Ce^{at} = |C|e^{rt}e^{j(\omega_0 t+\theta)}$$
$$= |C|e^{rt}[\cos(\omega_0 t + \theta) + j\sin(\omega_0 t + \theta)]$$

  - The envelope is $|C|e^{rt}$

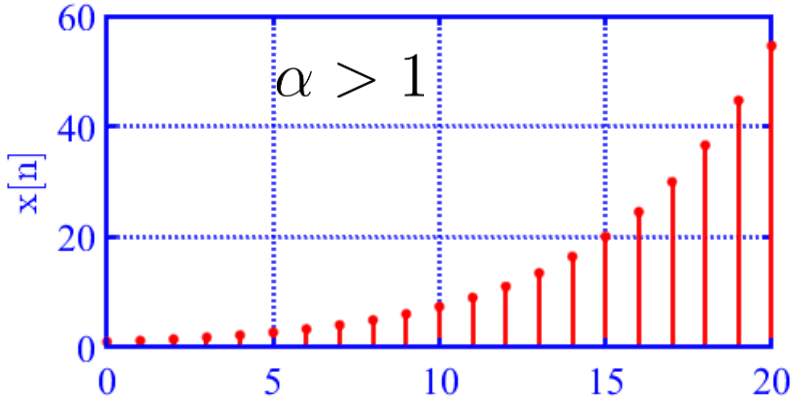


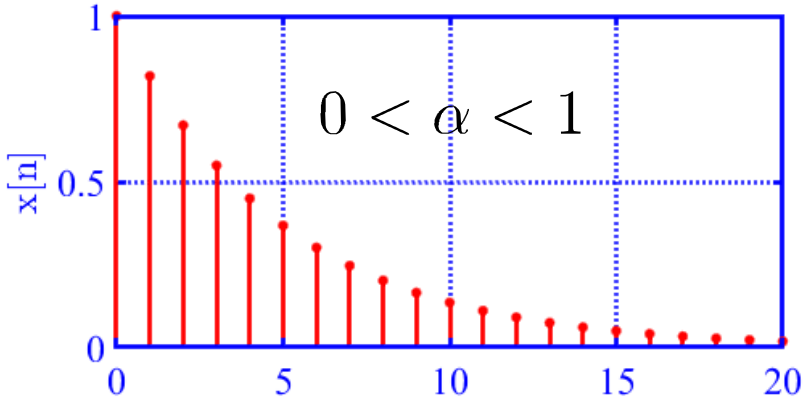
$r > 0$ $r < 0$ $r = 0$

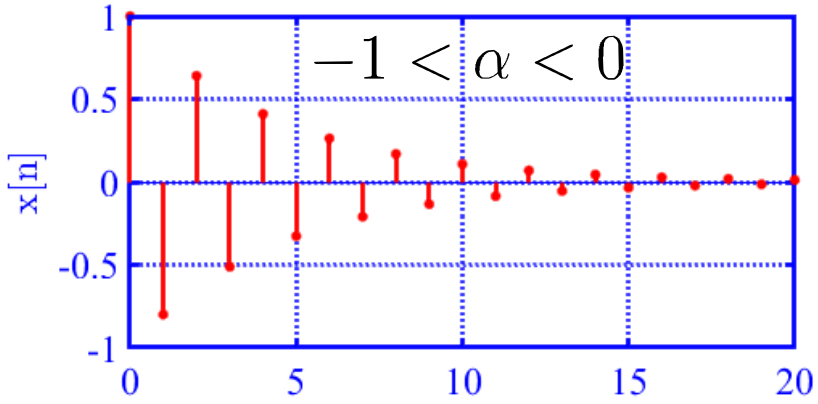
Most physical world signals

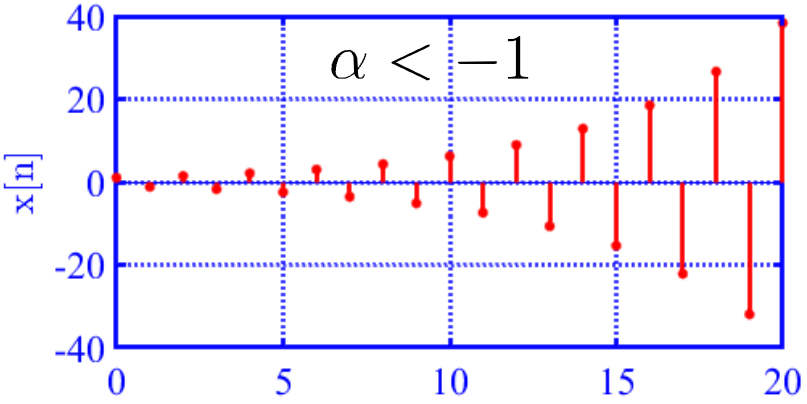
Discrete-Time Exponential and Sinusoidal Signals

- General form: $x[n] = C\alpha^n$ tradition
- Real exponential signals: When C and α are real









- Complex exponential signals: $x[n] = e^{j\omega_0 n}$
  - Euler's relation:

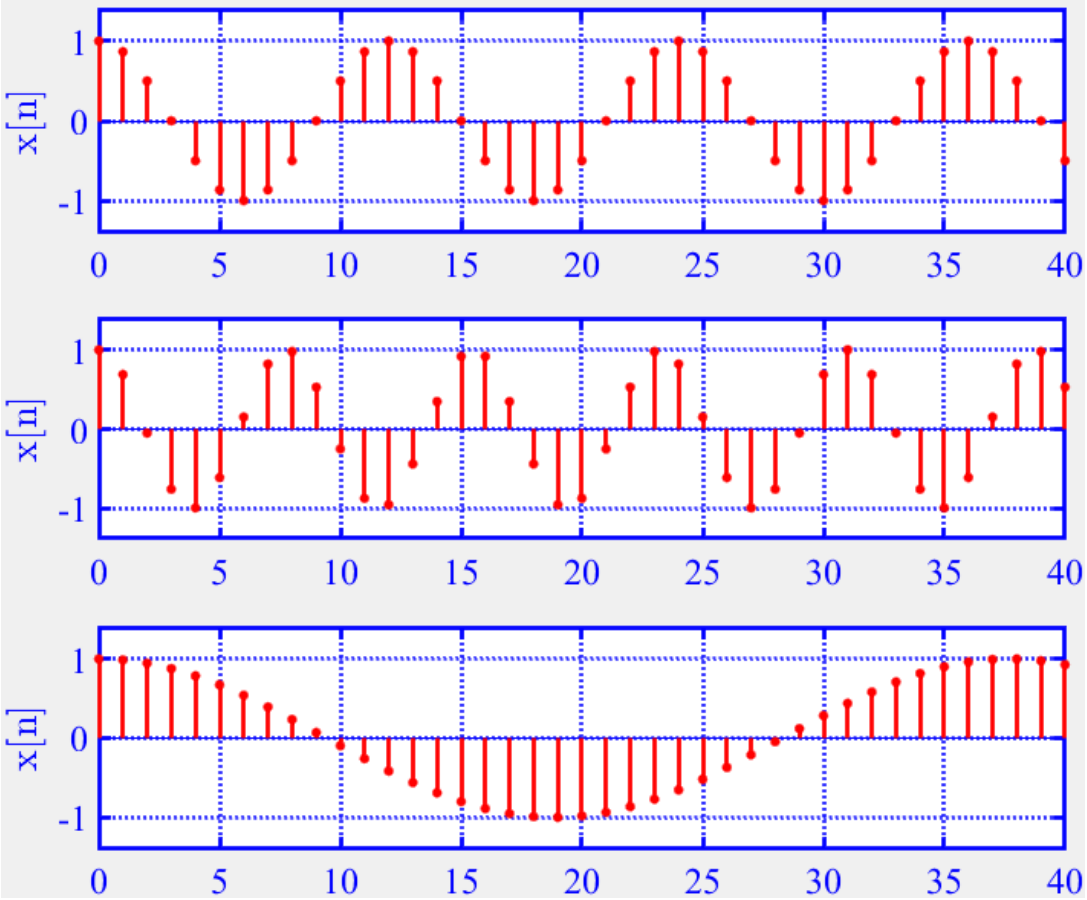
$$e^{j\omega_0 n} = \cos\omega_0 n + j\sin\omega_0 n$$

  - Power is just 1
  - Periodic?

$$x[n] = \cos(2\pi n/12)$$

$$x[n] = \cos(8\pi n/31)$$

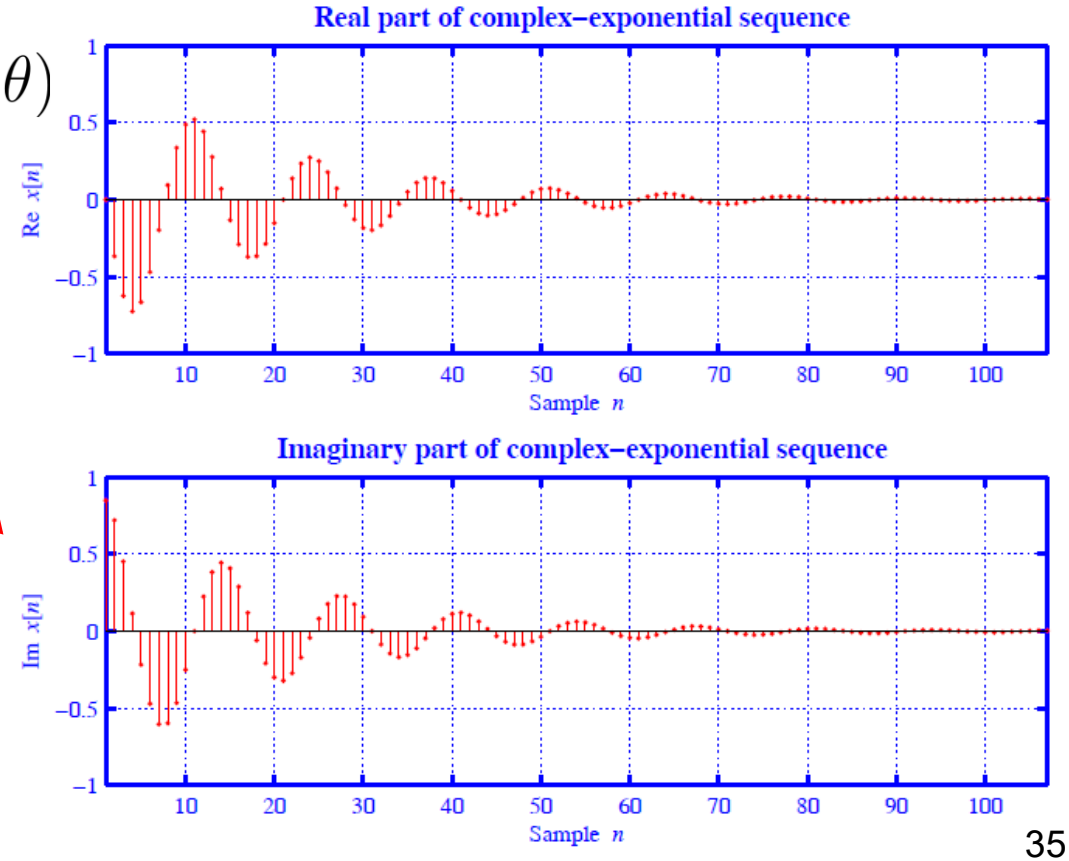
$$x[n] = \cos(n/6)$$



- General complex exponential signals:
  - Let $C = |C|e^{j\theta},\ \alpha = |\alpha|e^{j\omega_0}$

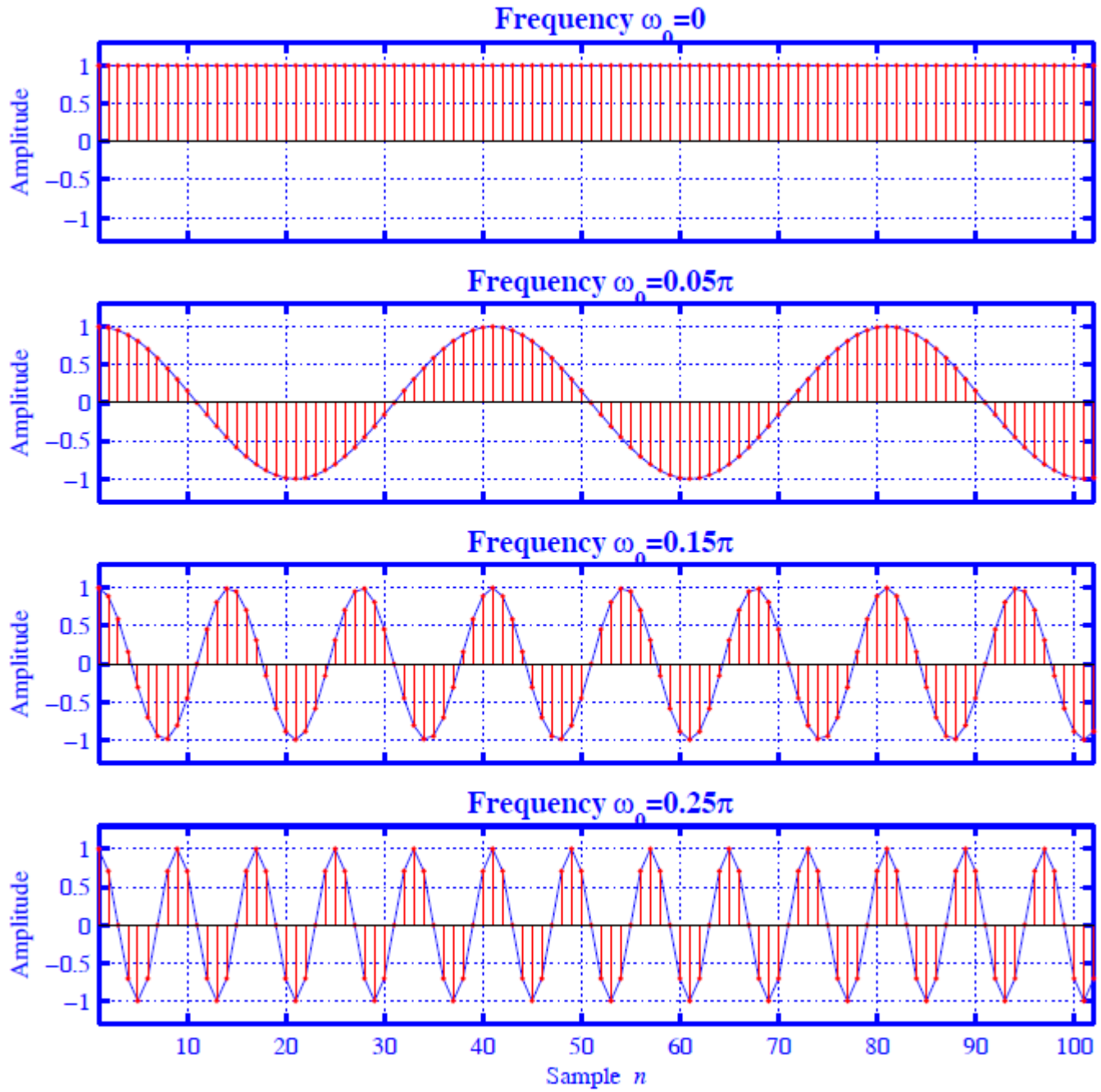
$$C\alpha^n = |C||\alpha|^n[\cos(\omega_0 n + \theta) + j\sin(\omega_0 n + \theta)]$$

  - $|\alpha| < 1$
  - $|\alpha| > 1$
  - $|\alpha| = 1$

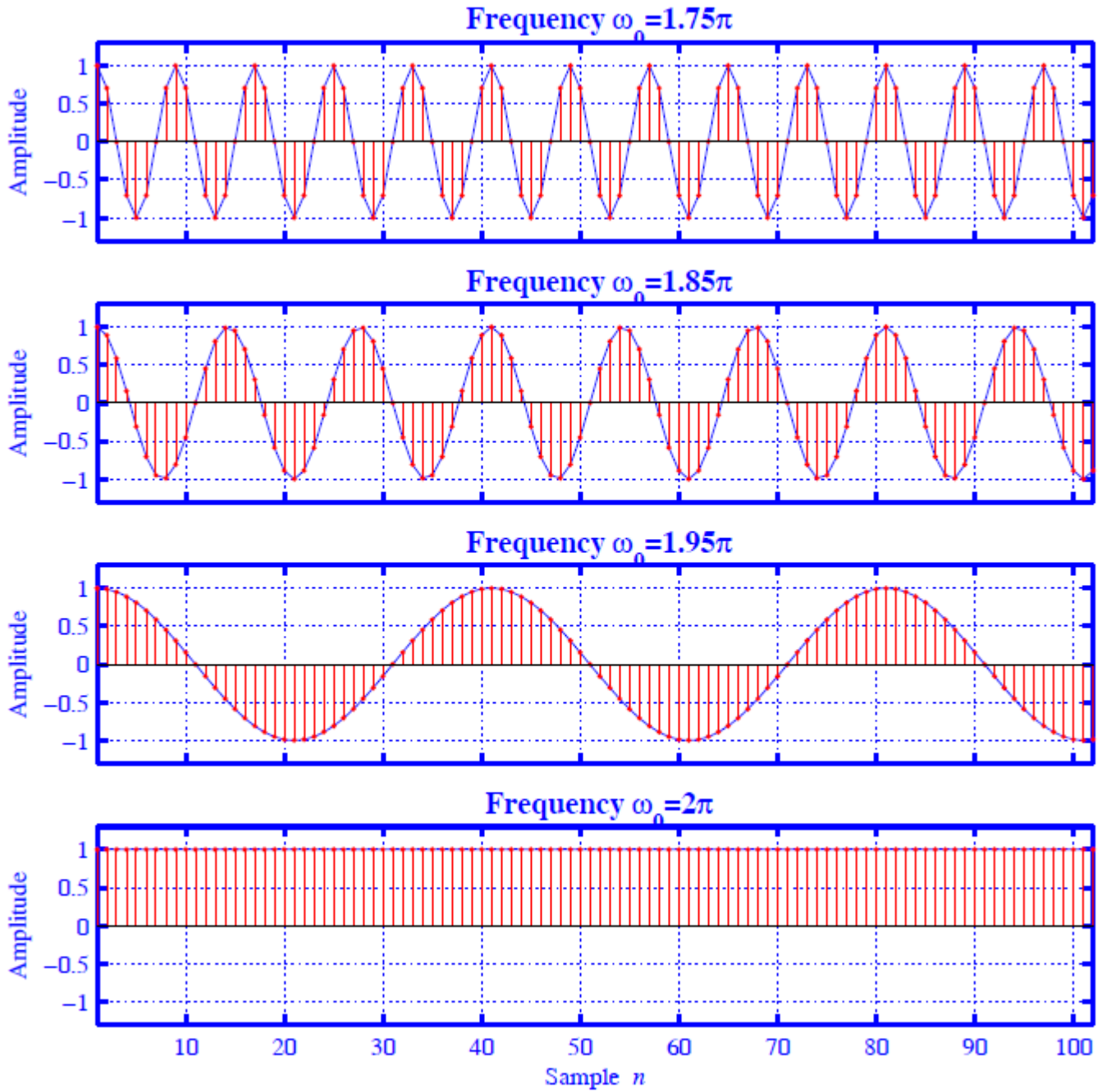


Periodicity: different from continuous-time partner.

- The rate of the oscillation is not proportional to ω_0
  - Since $e^{j\omega_0 n} = e^{j(\omega_0+2\pi)n}$, $e^{j\omega_0 n}$ is the same as $e^{j(\omega_0+2\pi)n}$. However in continuous case, $e^{j\omega_0 t}$ are different for different ω_0 (the frequency continuously increases with ω_0).
  - In discrete case we only need to consider $0 \le \omega_0 < 2\pi$, or $-\pi \le \omega_0 < \pi$ Periodicity in ω_0 (how about in n ?)
  - The rate of the oscillation is the lowest when $\omega_0 = 0$, while is the highest when $\omega_0 = \pi$, e.g., high frequency


Frequency ω0=0
Frequency ω0=0.05π
Frequency ω0=0.15π
Frequency ω0=0.25π
Amplitude
1
0.5
0
−0.5
−1
10
20
30
40
50
60
70
80
90
100
Sample n

$\cos \omega_0 n$



What happens when $\omega_0 = \pi$

- Not all $e^{j\omega_0 n}$ are periodical （in n）

$$e^{j\omega_0 n} = e^{j\omega_0 (n+N)} \Rightarrow e^{j\omega_0 N} = 1$$

$$\Rightarrow \omega_0 N = 2m\pi \quad \text{or} \quad \omega_0 = \frac{m}{N} 2\pi$$

ω_0 should be rational number multiply 2π

  - $e^{j\omega_0 n}$ is periodical only if $\omega_0/(2\pi)$ is a rational number.
  - If m and N has no factor in common, i.e., N is the smallest, then the latter is the fundamental period $N = 2m\pi/\omega_0$. Consequently, the fundamental frequency is

$$2\pi/N = \omega_0/m \leq \omega_0$$

The same holds for discrete sinusoid

  - Example---check the fundamental period and frequency of

$$\cos \pi n/6, \cos 8\pi n/31$$

N time domain points cover m 2π

– Example: determine the fundamental period of

$$e^{(j2\pi/3)n} + e^{(j3\pi/4)n}$$

- Harmonically related periodic exponentials with a common period N

$$\phi_k[n] = e^{jk(2\pi/N)n}, \quad k = 0, \pm 1, \pm 2, \ldots$$

Different from the continuous case, finite number here:

$$\phi_{k+N}[n] = \phi_k[n]$$

So, only N distinct harmonics exist:

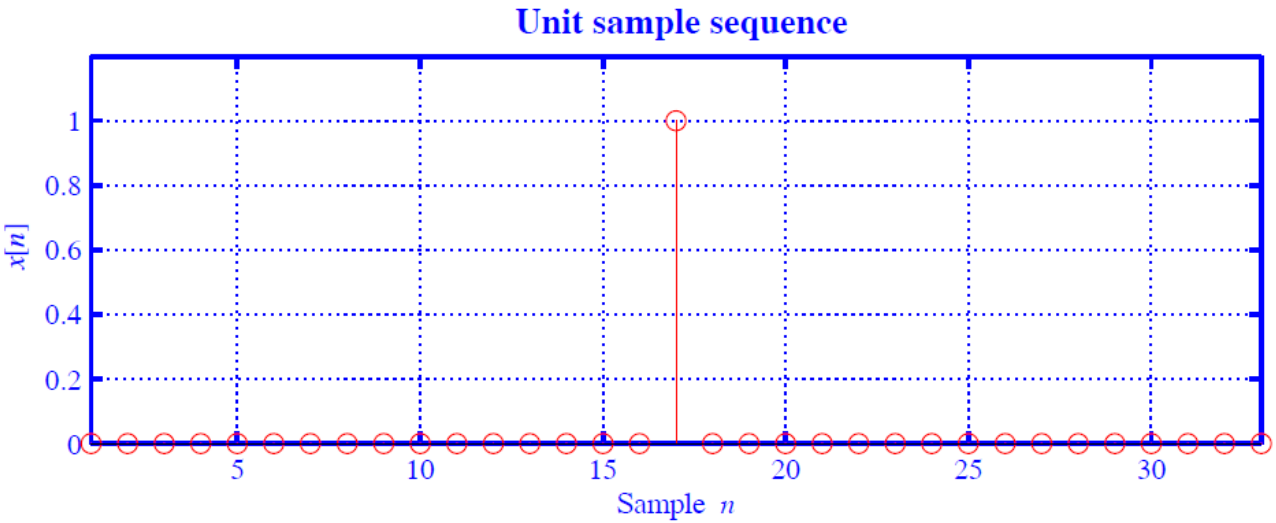
$$\phi_0[n] = 1, \phi_1[n] = e^{j2\pi n/N}, \ldots, \phi_{N-1}[n]$$

Different from CT case, $e^{j8\pi n/31}$ is the 4th component in the harmonic exponentials with period 31

Unit Impulse and Unit Step Functions

- Discrete unit impulse $\delta[n] = \begin{cases} 0 & n \neq 0 \\ 1 & n = 0 \end{cases}$

Unit sample/
Kronecker delta function



  - Important property (sampling):

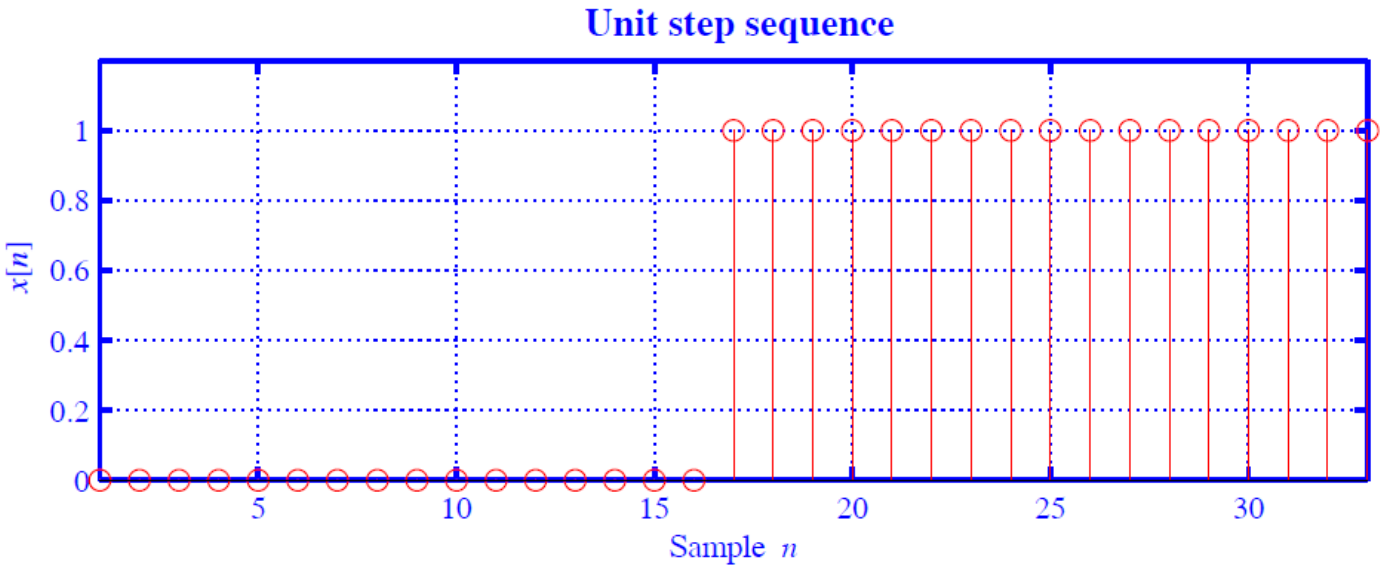
$$x[n]\delta[n] = x[0]\delta[n] \qquad x[n]\delta[n-k] = x[k]\delta[n-k]$$

  - More generally:

$$x[n] = \sum_{k=-\infty}^{\infty} x[k]\delta[n-k]$$

Unit Impulse and Unit Step Functions

- Discrete unit step: $$u[n] = \begin{cases} 0 & n \neq 0 \\ 1 & n \geq 0 \end{cases}$$



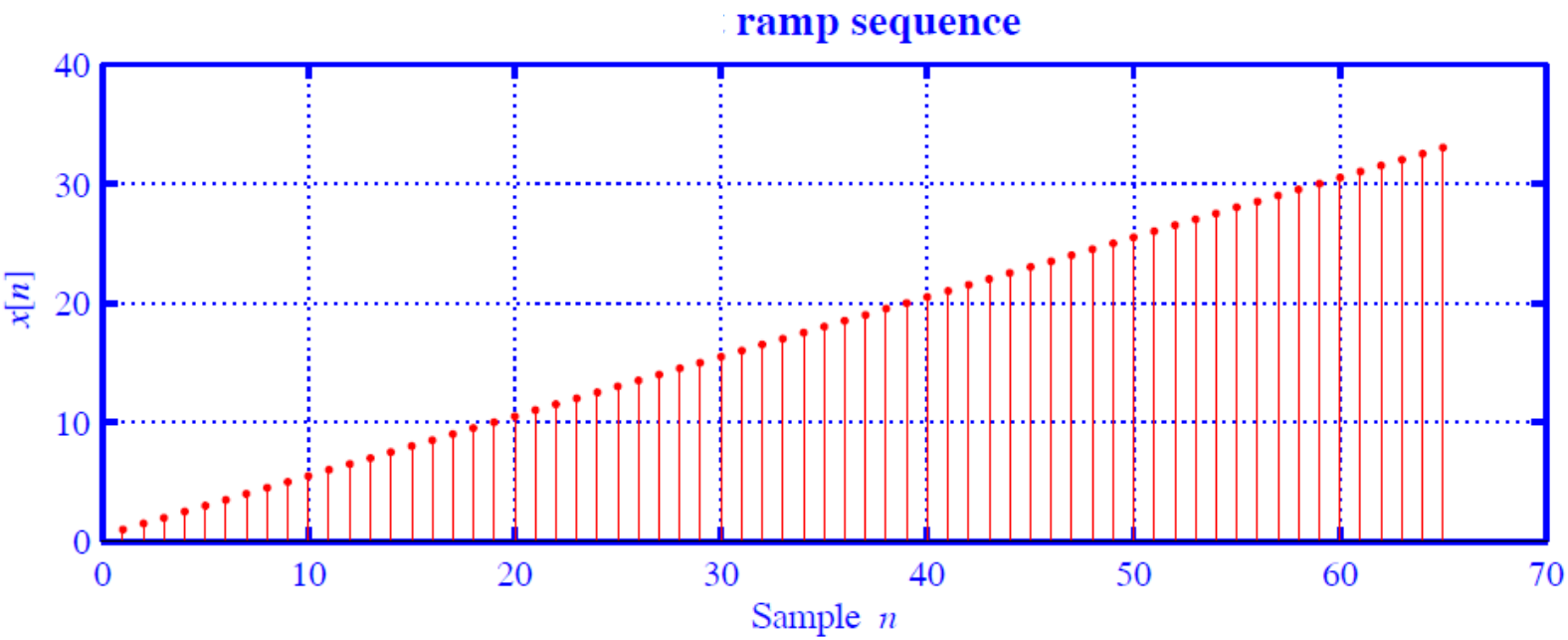
  - Relationship $\delta[n] = u[n] - u[n-1]$ first order difference

$$u[n] = \sum_{m=-\infty}^{n} \delta[m] = \sum_{k=0}^{\infty} \delta[n-k]$$ running sum

Unit Impulse and Unit Step Functions

- Discrete unit ramp

$$r[n] = \begin{cases} 0 & n \neq 0 \\ n & n \geq 0 \end{cases}$$



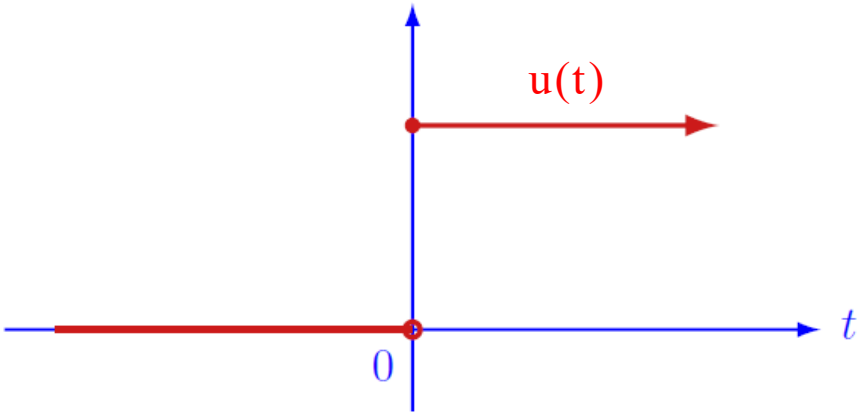
  - Relationship

$$r[n] = nu[n]$$

Unit Impulse and Unit Step Functions

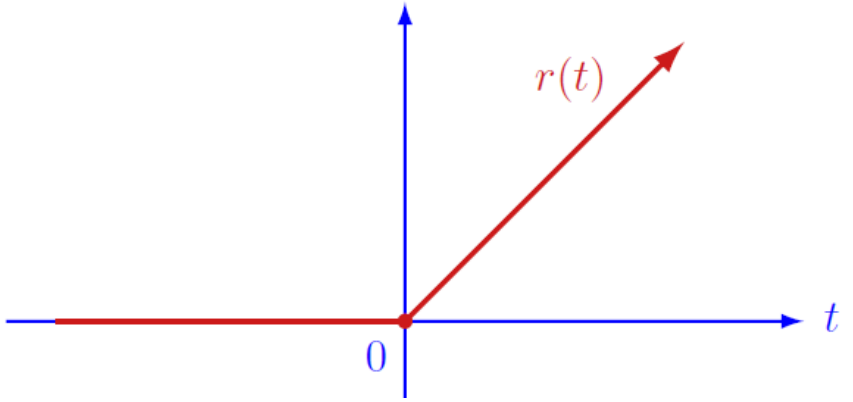
- Continuous unit step

$$u(t) = \begin{cases} 0 & t < 0 \\ 1 & t > 0 \end{cases}$$



- Continuous unit ramp

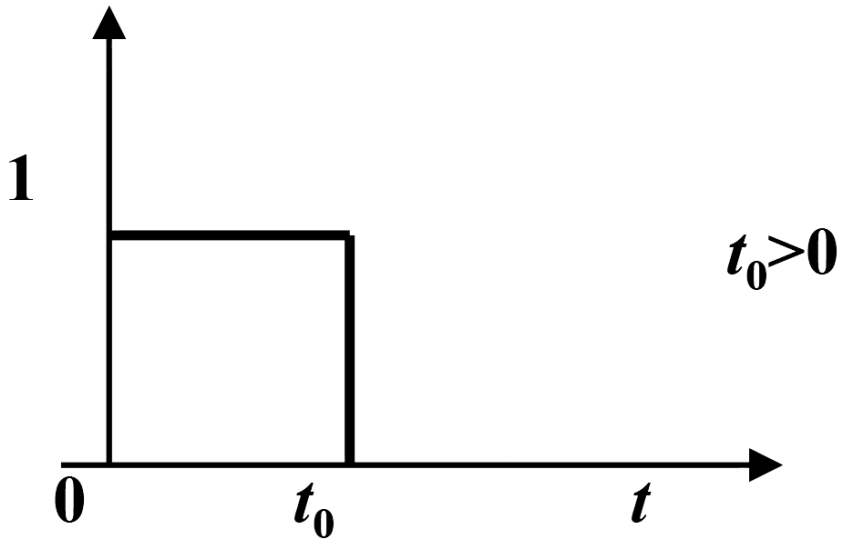
$$r(t) = \begin{cases} 0 & t < 0 \\ t & t \geq 0 \end{cases}$$

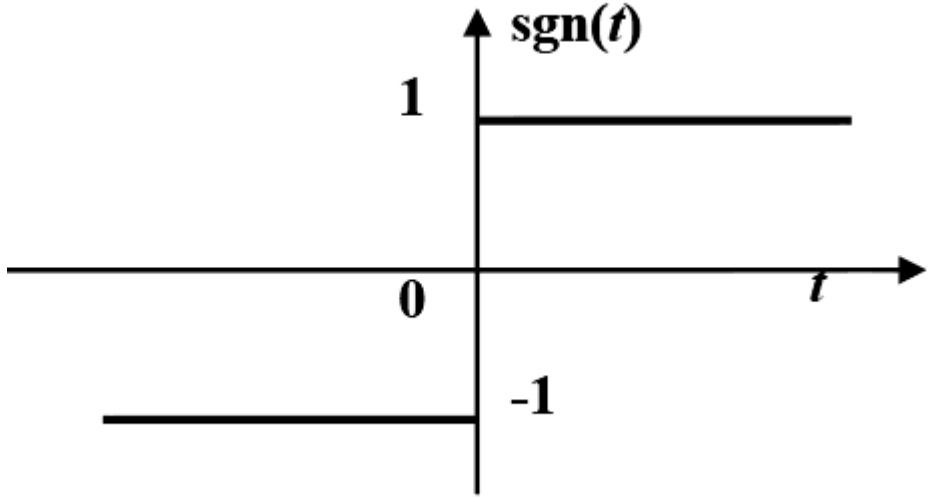


- Relationship

$$r(t) = \int_{-\infty}^{t} u(x)dx = tu(t) \qquad u(t) = \frac{dr(t)}{dt}$$

- Q1: Represent the rectangular pulse and sign function by unit step function



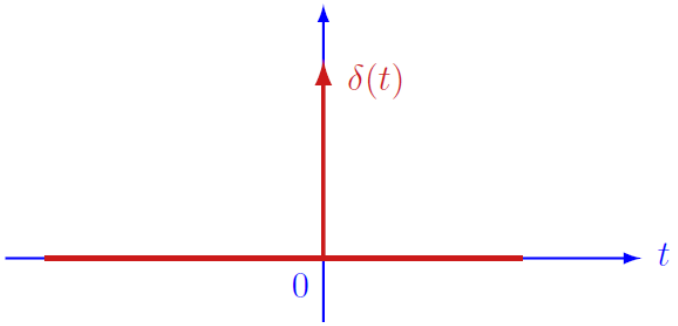


- Continuous unit impulse function should satisfy

$$\begin{cases} \delta(t) = 0, & t \neq 0 \\ \int_{-\infty}^{\infty} \delta(t)dt = 1 \end{cases}$$

  - It is useful to imagine $\delta(0) = +\infty$
  - Also named as Dirac Delta
  - By definition, there are

$$u(t) = \int_{-\infty}^{t} \delta(\tau)d\tau \qquad \delta(t) = \frac{du(t)}{dt}$$



No duration but unit area

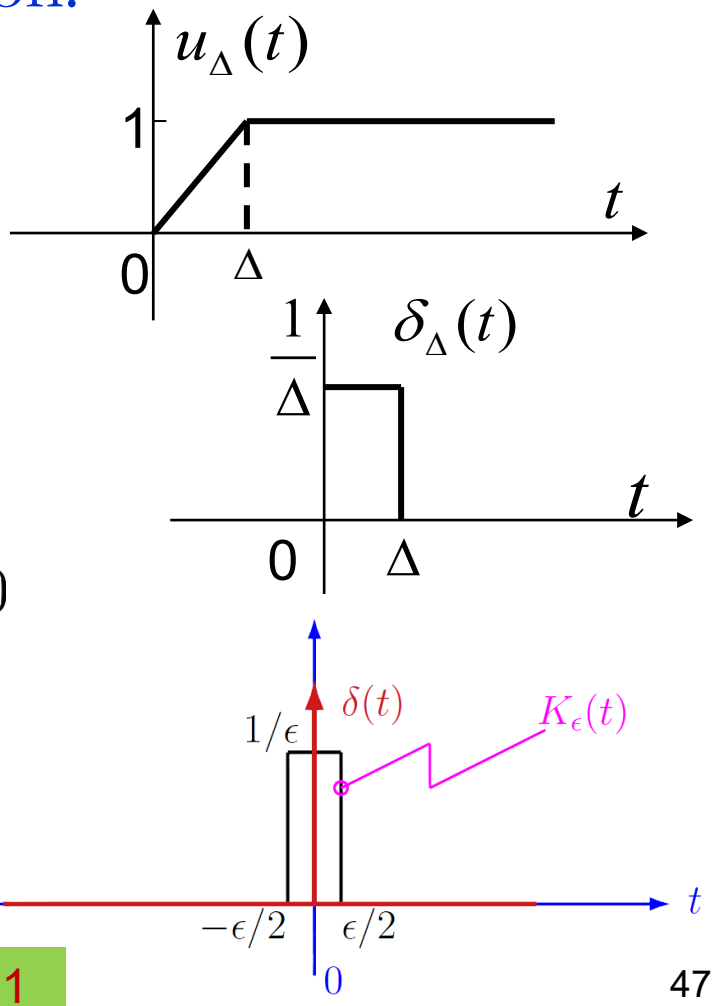
Dirac refers to the Quantum Physics pioneer Paul Dirac who used this to represent electrons as units of charge occupying an infinitesimal amount of space. Dirac delta function is one of the most important functions for understanding systems, but ironically, it does not exist in practice! It is "a really narrow, really tall pulse that has unit area".

- Understanding unit impulse function:
- Define $u_\Delta(t)$ as
  - $u_\Delta(t) \to u(t)$ when $\Delta \to 0$
- Define

$$\delta_\Delta(t) = \frac{du_\Delta(t)}{dt}$$

  - The area is always 1
  - Clearly $\delta_\Delta(t) \to \delta(t)$ when $\Delta \to 0$
- Alternatively, define

$$K_\epsilon(t) = \begin{cases} 1/\epsilon, & |t| \le \epsilon/2 \\ 0, & |t| > \epsilon/2 \end{cases}$$



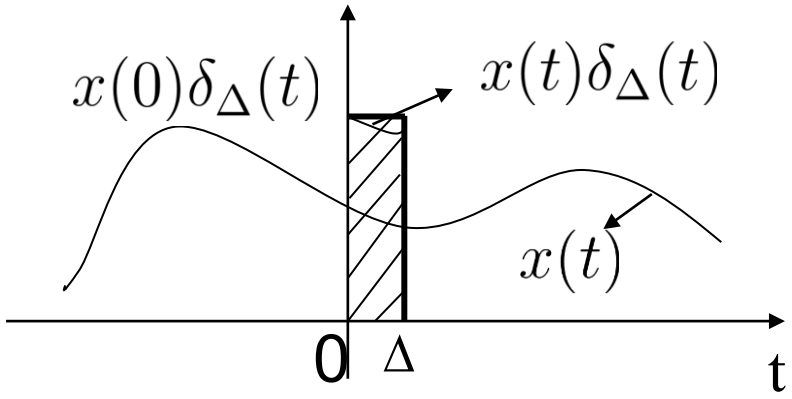
There are many other definitions as long as the area is 1

- Sampling Property:
  - There is $x(t)\delta_\Delta(t) \approx x(0)\delta_\Delta(t)$
  - When $\Delta \to 0$

    $$x(t)\delta(t) = x(0)\delta(t)$$
  - Similarly

    $$x(t)\delta(t-t_0) = x(t_0)\delta(t-t_0)$$
  - Hence

    $$\int_{-\infty}^{+\infty} x(t)\delta(t-t_0)dt = x(t_0)$$
  - Q2: what are $\delta(-t),\ \delta(2t)$?



Hence, $\delta(t-t_0)$ picks up the value of $x(t)$ at specific $t = t_0$. That means $\delta(t-t_0)$ is a distribution rather than function

Triangle pulse $\delta(t) = \lim_{\tau \to 0} \left\{ \frac{1}{\tau} \left(1 - \frac{|t|}{\tau} \right) [u(t+\tau) - u(t-\tau)] \right\}$

Two-sided exponential pulse $\delta(t) = \lim_{\tau \to 0} \left(\frac{1}{2\tau} e^{-\frac{|t|}{\tau}} \right)$

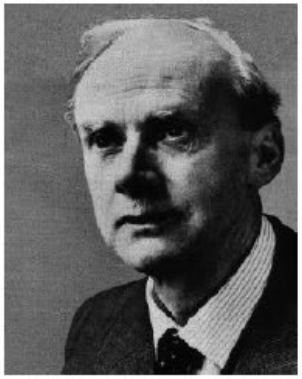
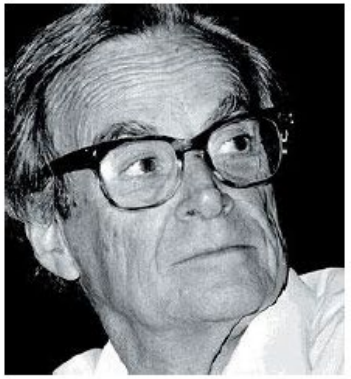
Gaussian pulse $\delta(t) = \lim_{\tau \to 0} \left(\frac{1}{\tau} e^{-\pi \left(\frac{t}{\tau} \right)^2} \right)$

And $x_a(t) = \frac{1}{\pi} \cdot \frac{a}{a^2 + t^2}$ $\delta(t) = \lim_{a \to 0} x_a(t)$

Can also define the area as E

Who is Who in This Lecturer

Paul A.M. Dirac——nobel-prize winning physicist (1933)

Paul Adrien Maurice Dirac (August 8, 1902~October 20, 1984) was a British theoretical physicist and a founder of the field of quantum mechanics. He held the Lucasian Professor of Mathematics at the University of Cambridge and spent the last ten years of his life at Florida State University. Among other discoveries, he formulated the so-called *Dirac equation* which describes the behavior of fermions and which led to the prediction of the existence of antimatter.

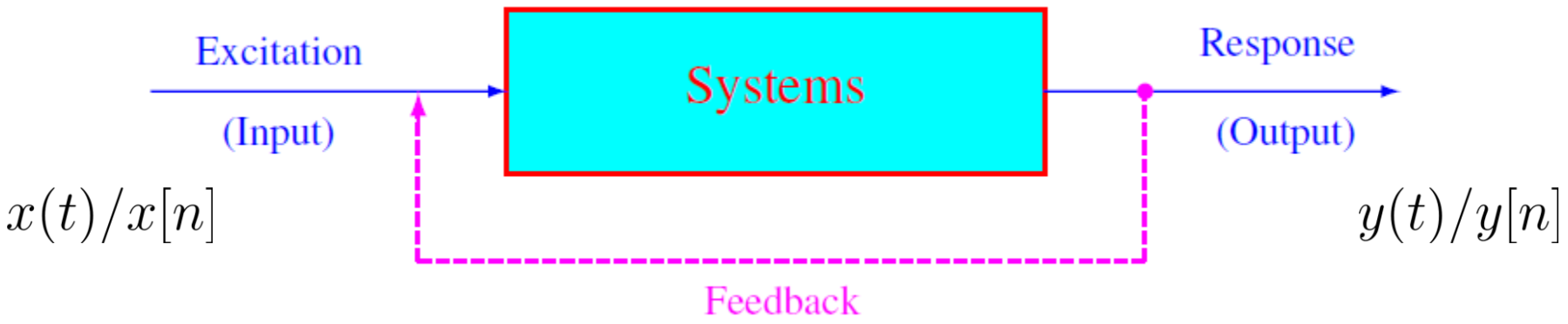
Honors and tributes

Dirac shared the 1933 Nobel Prize for physics in with Erwin Schrödinger "for the discovery of new productive forms of atomic theory." Dirac was also awarded the Royal Medal in 1939 and both the Copley Medal and the Max Planck medal in 1952.

He was elected a Fellow of the Royal Society in 1930, and of the American Physical Society in 1948.

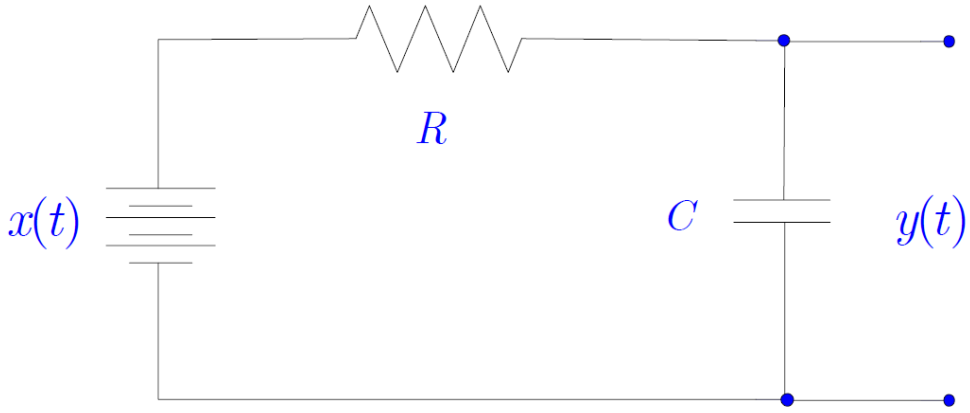
Systems

- Physically, a system is something that takes in one or more input signals and produces one or more output signals according to certain rules
  - Maybe it is a circuit
  - Maybe it is a mechanical thing
  - Maybe it is ?

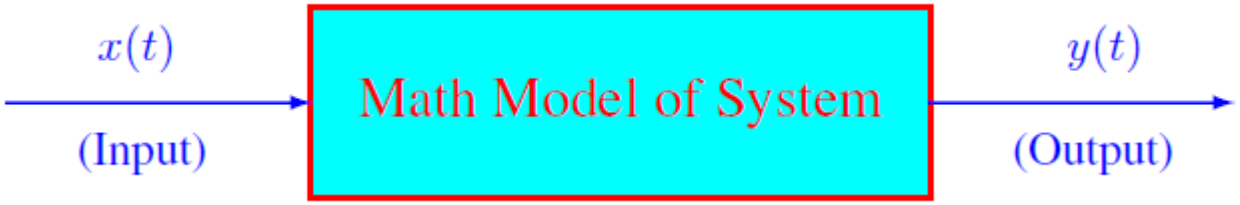


- Physical view and Schematic view of a system





- The math model gives a way to quantitatively relate an input signal into the output signal

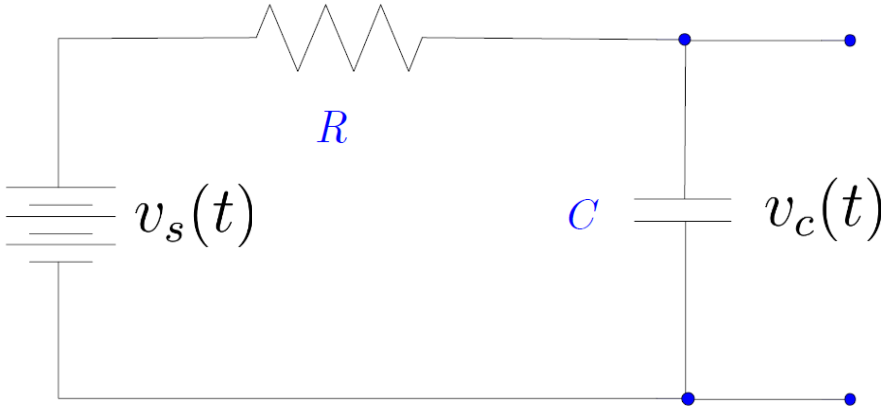


Continuous-Time Systems

- Continuous systems $x(t) \rightarrow y(t)$
  - Example: input $v_s(t)$, output $v_c(t)$

$$i(t) = \frac{v_s(t) - v_c(t)}{R}$$

$$i(t) = C\frac{dv_c(t)}{dt}$$



$$\Longrightarrow \frac{dv_c(t)}{dt} + \frac{1}{RC}v_c(t) = \frac{1}{RC}v_s(t)$$

Linear Differential Equations

– Example: input $f(t)$ output $v(t)$

$$\frac{dv(t)}{dt} = \frac{1}{m}[f(t) - \rho v(t)]$$

$$\Longrightarrow \frac{dv(t)}{dt} + \frac{\rho}{m}v(t) = \frac{1}{m}f(t)$$

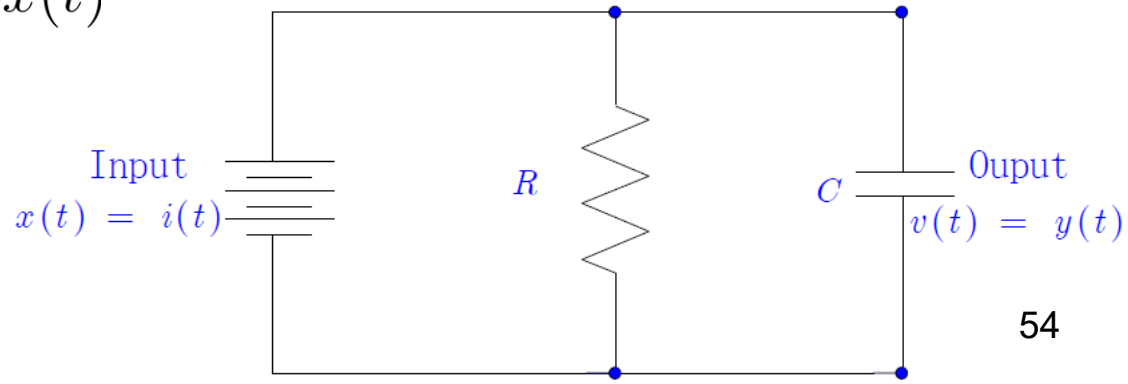


- A common structure: linear differential equations

$$\frac{dy(t)}{dt} + ay(t) = bx(t)$$

- Q: input $i(t)$

  output $v(t)$



- Discrete systems $x[n] \rightarrow y[n]$
  - Example: balance in a bank, 1% monthly interest

$$y[n] = 1.01y[n-1] + x[n]$$

  - Population in one country, death rate k_d , birth rate k_b

$$y[n] = (1 + k_b - k_d)y[n-1] + x[n]$$

- A common structure: difference equation

$$y[n] + ay[n-1] = bx[n]$$

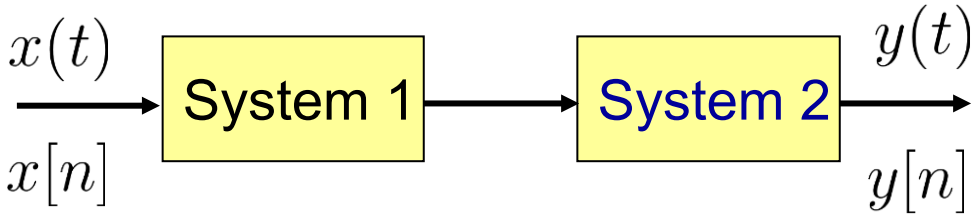
Continuous-time systems are generally described by differential equations;
Discrete-time systems are generally described by difference equations.

Remark

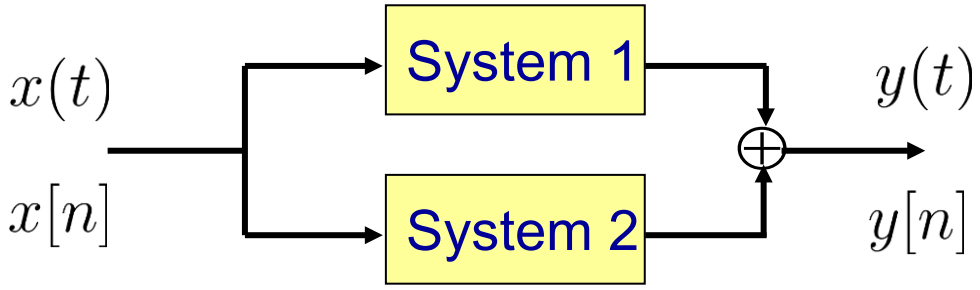
- For various applications, the mathematical modeling of the systems would be amazingly similar.
- Mathematical modeling would be an abstract of the practical system which may not necessarily accurate.
- One need to identify the valid region of the mathematical modeling for the system, and design the latter algorithm within this valid region.
- What happens if the modeling fails?

Interconnection of Systems

- To build a complicated system with sub-systems
  - Series or cascade interconnection

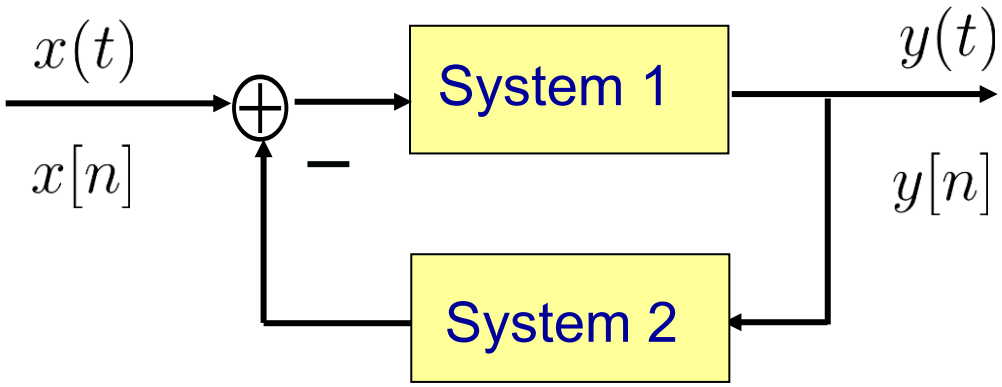


  - Parallel interconnection:

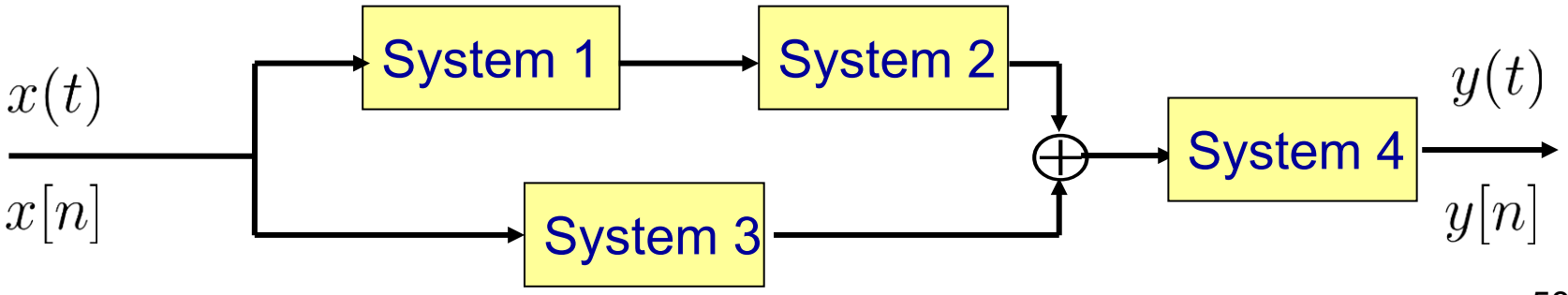


Block Diagram

– Feedback interconnection



– Series-parallel interconnection



Basic System Properties

- Memoryless and memory system
  - Output at a given time only depends on the current input

$$y(t) = 2x(t)$$ amplifier

  - Otherwise:

$$y[n] = x[n - n_0]$$ shifter

$$y[n] = \sum_{k=-\infty}^{n} x[k]$$ accumulator

$$y(t) = \frac{1}{C}\int_{-\infty}^{t} x(\tau)d\tau$$ capacitor

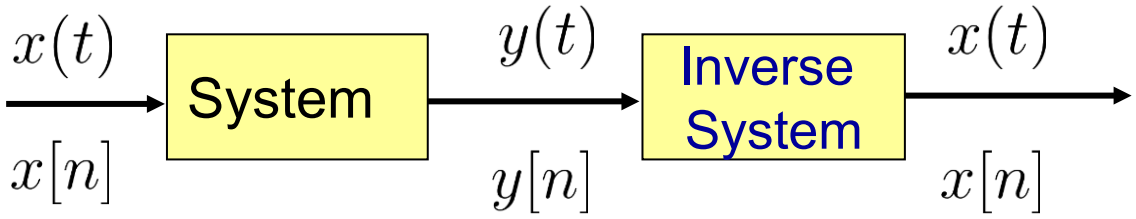
Roughly, memory system normally has the mechanism to store the information of the input signals (past/future)

cost and energy

$$y[n] = \sum_{k=-\infty}^{n} x[k] = y[n-1] + x[n]$$

Invertibility and Inverse System

– Distinct inputs lead to distinct outputs, cascade yields original



$$y(t) = Rx(t) \quad \Rightarrow \quad x(t) = y(t)/R$$

$$y[n] = \sum_{k=-\infty}^{n} x[k] \quad \Rightarrow \quad x[n] = y[n] - y[n-1]$$

$$y(t) = \frac{1}{C}\int_{-\infty}^{t} x(\tau)d\tau \quad \Rightarrow \quad x(t) = C\frac{dy(t)}{dt}$$

– Otherwise

$$y(t) = x^2(t) \qquad y(t) = 0 \qquad y[n] = x[2n]$$

Causality

- A system is causal if the output at any time only depends on the present and the past inputs.

  Amplifier, accumulator, shifter for $n_0 \geq 0$

  $$y(t) = g(t)x(t)$$ multiplier

  – Otherwise:

  $$y[n] = x[n - n_0], \; n_0 < 0$$ shifter

  $$y[n] = \frac{M}{2M+1} \sum_{k=-M}^{M} x[n-k]$$

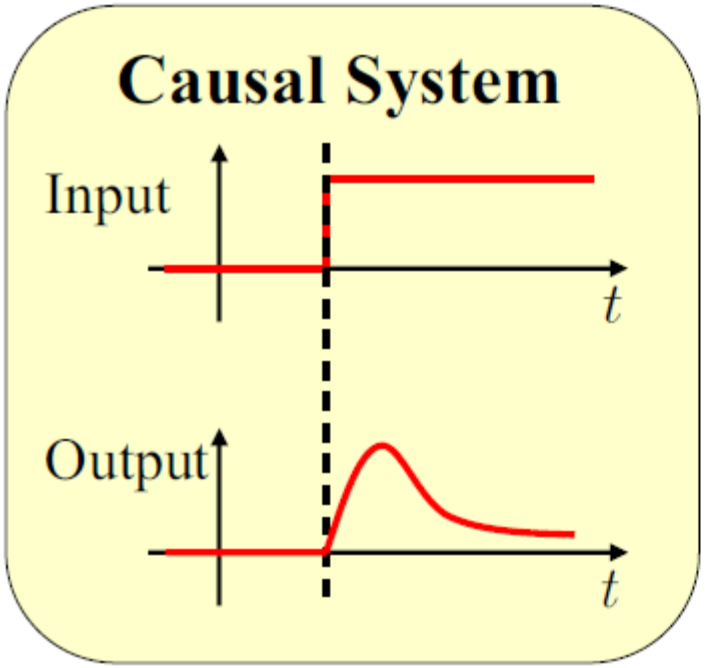
  $$y[n] = x[-n]$$

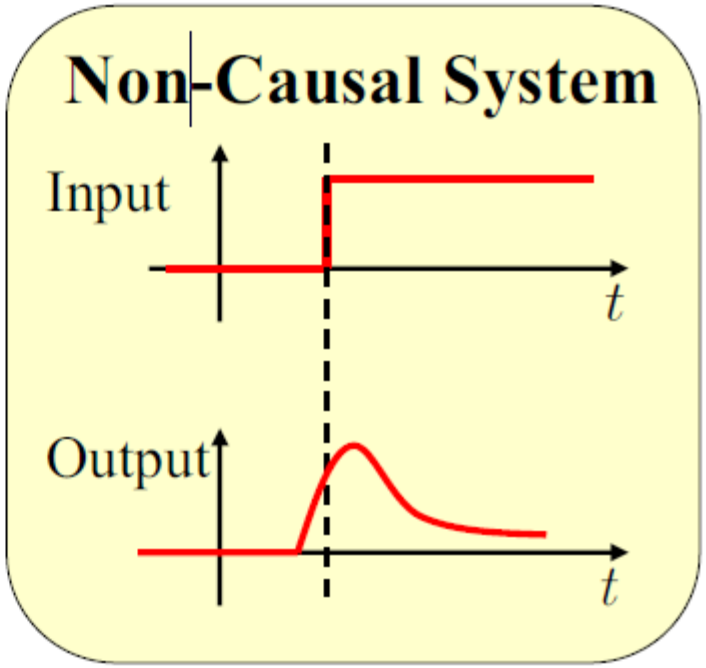
  – Physical world online systems are causal

  – Memoryless systems are causal

How about
$y(t) = x(t)\cos(t+1)$

– A Causal system (with zero initial conditions) cannot have a non-zero output until a non-zero input comes





– But we need to understand non-causal systems because theory shows that the "best" systems are non-causal!

Stability

- Formal definition: if the input to a stable system is bounded, then the output must also be bounded
  - A stable system is the one in which small inputs lead to outputs that do not diverge.
  - Naturally existed systems, e.g., RLC circuit, due to energy loss
  - How about bank account accumulation?
- Mathematically, we can assume $|x(t)| \leq B$, and check whether $|y(t)| \leq C$ holds for some C.

$$y[n] = \sum_{k=-\infty}^{n} x[k] \qquad y[n] = \frac{M}{2M+1} \sum_{k=-M}^{M} x[n-k]$$

$$y(t) = tx(t) \qquad y(t) = e^{x(t)}$$

Time Invariance

- A system's behavior does not change with time, i.e., today' experiment is the same as tomorrow's for the same input.
  - Specifically, a time shift in the input signal results in an identical time shift in output signals

$$x(t-t_0) \rightarrow y(t-t_0) \qquad x[n-n_0] \rightarrow y[n-n_0]$$

  - Example:

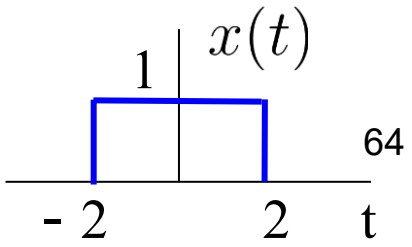
$$y(t) = 2x(t) \qquad y(t) = \sin[x(t)]$$

  - Example:

$$y[n] = nx[n] \qquad y(t) = x(2t)$$

Counterexample: $x[n] = 1$

Counterexample

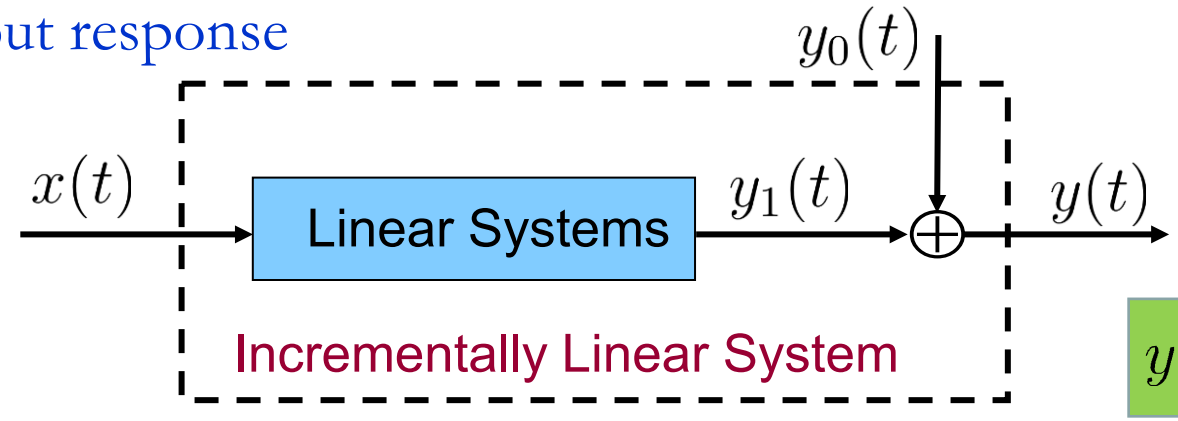


Linearity

- The system is linear if
  - The response to $x_1(t) + x_2(t)$ is $y_1(t) + y_2(t)$ additivity
  - The response to $ax(t)$ is $ay(t)$ (complex a) scaling
  - The response to $ax_1(t) + bx_2(t)$ is $ay_1(t) + by_2(t)$ superposition
- Example
  - Zero input $x(t) = 0$
  - $y(t) = tx(t)$ $y(t) = x^2(t)$ $y[n] = \Re\{x[n]\}$
  - $y[n] = 2x[n] + 3$ Linear equation but not linear system

When superposition holds true, it makes our life easier! We then can decompose complicated signals into a sum of simpler signals and then find out how each of these simple signals goes through the system!

- Incrementally linear systems
  - The difference between the responses is linear to the difference between the two inputs
  - Superposition of the response of a linear system with a zero-input response



$$y[n] = 2x[n] + 3$$

  - Zero-input response: when the input of the system is zero, $y_0(t)$
  - Zero-state response: when the initial condition of the system is zero, $y_1(t)$

Summary

- The Mathematical modeling of the signals and systems
- The operations of signals: shift/time-reversal/scaling
- The basic continuous-time and discrete-times signals exponentials, sinusoids, unit-impulse, unit step
- Periodicity of continuous-time and discrete-time signals
- Six properties of the systems as well as the incrementally linear system

Most systems can be modeled as linear time-invariant (LTI) systems, based on which we can analyze the systems completely and universally

Homework #1

- 1.9(b)(c)(e), 1.10, 1.15, 1.16, 1.17, 1.21(a)(d)(f), 1.25(b)(c)(e), 1.27(a)(c)(d)(g), 1.29, 1.31,
- 1.33, 1.36, 1.41, 1.42
- Try whenever you think you can handle.
- Try finish all questions after the chapter.
- Homework is always due 1 week after each Chapter is finished.